



Parameter estimation from single patient, single time-point sequencing data of recurrent tumors

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Abstract

In this study, we develop consistent estimators for key parameters that govern the dynamics of tumor cell populations when subjected to pharmacological treatments. While these treatments often lead to an initial reduction in the abundance of drug-sensitive cells, a population of drug-resistant cells frequently emerges over time, resulting in cancer recurrence. Samples from recurrent tumors present as an invaluable data source that can offer crucial insights into the ability of cancer cells to adapt and withstand treatment interventions. To effectively utilize the data obtained from recurrent tumors, we derive several large number limit theorems, specifically focusing on the metrics that quantify the clonal diversity of cancer cell populations at the time of cancer recurrence. These theorems then serve as the foundation for constructing our estimators. A distinguishing feature of our approach is that our estimators only require a single time-point sequencing data from a single tumor, thereby enhancing the practicality of our approach and enabling the understanding of cancer recurrence at the individual level.

Keywords Cancer recurrence · Branching process · Parameter estimation

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1 Introduction

A significant obstacle in the effective treatment of cancer is the ability of cancer cells to develop resistance to pharmacological treatments. Drug resistance in cancer often

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stems from genetic mutations (single nucleotide variants or copy number alternations), which can modify the structure of proteins targeted by anticancer drugs, activate alternate signaling pathways circumventing the drug's action, or enhance the cancer cells' internal repair mechanisms (Housman 2014). Mutated cells, endowed with such drug-resistant traits, gain a selective advantage over their drug-sensitive counterparts under treatment pressures. This clonal selection fosters the proliferation of drug-resistant cancer cells bearing diverse mutations, resulting in significant heterogeneity within recurrent tumors (Ding et al. 2012; Hunter 2006). While such intratumor heterogeneity within recurrent tumors presents many challenges for subsequent therapy, it also offers an unique opportunity to decode the dynamics of cancer recurrence by examining the timing of occurrence and the tumor's clonal structure at cancer recurrence (Leder and Wang 2021).

High-throughput sequencing has enabled the reconstruction of the subclonal architecture of tumor samples. Initially, genomic sequencing profiles allow the accurate identification of somatic mutations in cancer cells, such as single nucleotide variants (SNVs) (Cibulskis et al. 2013) and copy number alterations (CNAs) (Ha et al. 2014). Subsequently, by clustering the cancer cell fractions associated with these mutations, indicating the proportion of cancer cells carrying each mutation, one can deduce the subclones present in the corresponding tumor sample (Dentro et al. 2017). Subclonal analysis methods utilize SNVs, CNAs, or a combination of both (cf. Roth et al. 2014 and VanLoo et al. 2010). The ability to detect smaller subclones is influenced by sequencing depth and sampling strategy. For instance, deep sequencing ($\geq 500X$) unveils finer subclones compared to shallower sequencing ($\leq 50X$) (Stead et al. 2013). Additionally, employing multiple tumor samples (or even single cells) for sampling provides a higher resolution than relying on a single sample (Miller et al. Aug 2014). The methods we propose would require knowledge of the cancer cell fractions at the time of cancer recurrence.

Recent research has delved deeply into the mathematical modeling and analysis of cancer recurrence. Foo and Leder (2013) presented a model that encompasses two types of cancer cells, investigating the dynamics of a subcritical branching process that avoids extinction. They identified two pivotal stochastic moments in cancer recurrence: the 'turn-around time' where the total cell population begins to grow again, and the 'crossover time' when resistant cells become dominant. Building on this foundation, Foo et al. (2013) expanded the model to account for heterogeneous escape populations, taking into account the variable fitness of resistant groups sourced from a mutational fitness landscape. This led to insights into the regrowth kinetics and the influence of distribution shape on recurrent tumor composition. Another study (Jasmine and Leder 2012) delved deeper, focusing on rare events that precipitate early tumor recurrence, particularly the likelihood that the resistant cell population exceeds its average at the crossover time. Further contributions to this discourse include a study (Foo et al. 2014) that evaluates both random and deterministic fitness of resistant cells, providing evidence for a weak limit on the crossover time in both scenarios. Li et al. (2023) investigated resistance mechanisms, both amplification-driven and mutation-driven. They established a law of large numbers that accounts for the convergence of the stochastic recurrence time across both mechanisms, defining this time as when the cancer cell population exceeds its original size. Avanzini and Antal (2019) delved into

the issue of cancer relapse due to potential metastases. They introduced the concept of ‘recurrence time’ as the stochastic moment when a metastasis first becomes detectable and derived the probability distribution for this occurrence. Taking the discourse further, Hanagal et al. (2022), Leder and Wang (2021) enriched the understanding of cancer recurrence. Their focus was on examining the large deviations property of recurrence time and presenting an accurate depiction of the expected clonal diversity at recurrence, with a particular emphasis on early recurrence scenarios. Notably, while these studies have offered significant insights into recurrence time and the associated clonal heterogeneity, there remains an uncharted territory. There is a lack of strong convergence results vital for establishing consistent estimators related to metrics that detail clonal diversity, such as the number of clones or the Simpson’s Index. This paper sets out to address this gap.

In the past decades, a growing body of research has been committed to developing estimators for parameters crucial in understanding tumor dynamics. Liang and Sha (2004) utilized a deterministic, biexponential non-linear mixed-effects model to capture the potential tumor recurrence under treatment. They used tumor volume data from a study on mouse xenograft tumors to estimate the decay rates of the tumor’s response to treatment. Lee and Bozic (2022) utilized a two-type branching process model to investigate the scenario where one or more initial wild-type cancer cells acquire an additional driver mutation as the tumor evolves. In their approach, they developed a novel methodology that leverages bulk sequencing data obtained at two separate time points, without any intervening treatment, to derive estimates for critical parameters in chronic lymphocytic leukemia. These include the net growth rate of the cancer cells and the timing of the appearance of the additional driver mutation. Werner et al. (2020) investigated a tumor population model characterized by exponential growth, employing coalescence theory to estimate the expected distribution of mutational distances from multi-sample sequencing data. In this context, mutational distances represent the mutation count disparities between two ancestral cells of different samples. Following this, they implemented the Metropolis-Hastings algorithm in conjunction with sequencing data from multiple samples at a single time point, enabling the estimation of both the mutation rate and cell survival rate throughout the expansion of the tumor population. Additional research efforts are currently delving into the estimation of parameters by leveraging data on mutation frequencies. Salichos et al. (2020) examined the variant-allele frequency of generational hitchhikers neutral mutations that arise prior to driver mutations. They used this data to infer the presence of subclonal drivers and estimate the fitness advantage conferred by these drivers. In a series of works (Gunnarsson et al. 2021, 2023), Gunnarsson and colleagues adopted the Galton–Watson branching process to precisely describe the law of large numbers for the site frequency spectrum of neutral mutations, considering both large time and size limits. Utilizing these results, they formulated estimators for the extinction probability and mutation rate in a birth-death process. Our study differs from existing literature in that, unlike most previous research which primarily focuses on the net growth rate of tumor cells, our approach allows for the estimation of both the birth and death rates of tumor cells. Additionally, our study is distinct in providing theoretical guarantees regarding the consistency and convergence rates of the estimators, a crucial aspect often neglected in the current body of literature.

Lastly, this work is related to the stream of literature that studies the clonal structure of tumors (see for examples Cheek and Antal 2020; Durrett et al. 2011a; Nicholson et al. 2023; Brouard 2023 and references therein). This stream of literature primarily seeks to characterize the sizes of all the mutant sub-populations in the large population limit. A focal point of these studies pertains to the mutation rates, wherein certain studies (e.g., Cheek and Antal 2020) consider the rare mutation limit such that the product of the mutation rate μ_n and the targeted population size n converges to a finite limit (i.e., $\lim_{n \rightarrow \infty} n\mu_n = c < \infty$), while other studies (e.g., Brouard 2023) consider the power law mutation rates limit such that $\mu_n \propto n^{-\alpha}$ with $\alpha \in (0, 1)$. Our study adopts the latter perspective that mutation rates at the individual cell level are negligible ($\mu_n \ll 1$), but at the population level, the cumulative mutation rate is substantial ($n\mu_n \gg 1$), resulting in a high probability of the emergence of drug-resistant cells.

In this work, we adopt a two-type branching process model. The process starts with n drug-sensitive tumor cells under pharmacological treatments (i.e., there is no pre-existing resistance before the treatment), evolving over time as the tumor develops resistance through mutation with associated fitness changes. It should be noted that much of the existing literature considers cases where resistant mutants exist before treatment begins (see for examples Bozic and Nowak 2014; Bozic et al. 2013; Iwasa et al. 2006; Komarova and Wodarz 2005). We do not consider such settings in this study and leave that for future work. We define the recurrence time as the first time that the population of resistant cells reaches the initial tumor burden, n . We provide a set of convergence in probability results for various quantities observed at cancer recurrence. These include the recurrence time itself, denoted by γ_n , the number of mutant clones present at recurrence, denoted by $I_n(\gamma_n)$, and the Simpson's Index of mutant clones at recurrence, denoted by $R_n(\gamma_n)$. The challenge in establishing these results arises from the intricate nature of the underlying stochastic process and the randomness of the recurrence time. Leveraging the convergence in probability results of these quantities, we construct estimators for the net growth rate of both sensitive and resistant cells, the birth rate of resistant cells, and the mutation rate. Compared to our previous work (Leder and Wang 2021) and other related studies in the field, a distinguishing feature of our approach is that our estimators only require a single time-point sequencing data from a single tumor. Additionally, our estimators exhibit statistical consistency, thereby enhancing their reliability in characterizing individual tumors, which aids in personalized medical recommendations.

The organization of this paper is as follows. In Sect. 2, we introduce our model and previous results. In Sect. 3, we present the theoretical results, including the convergence in probability results for various quantities at cancer recurrence, and the consistency of our proposed estimators. In Sect. 4, we evaluate the performance of the estimators through simulation experiments. Lastly, we discuss the limitations of our approach and possible directions for further research in Sect. 5.

2 Model and previous results

2.1 Basic model underlying the recurrence process

We assume there are two types of cancer cells: sensitive cells and resistant cells. At the time $t = 0$, the tumor is comprised solely of sensitive cells with an initial population size denoted by n , which we assume to be a known parameter. Typically, n is an extremely large integer (a tumor with a volume of 1 cm^3 is commonly assumed to contain 10^9 cells). A pharmacological treatment is applied at the time $t = 0$, which causes the number of sensitive cells to decrease over time.

Remark 2.1 It is important to note that the emergence of drug-resistant cells can occur before the initiation of pharmacological treatment, which suggests an alternative model that assumes a non-zero initial population of resistant cells (Hanagal et al. 2022). We plan to investigate this phenomena in future work.

Here, we denote by $Z_0^n(t)$ the number of sensitive cells at time t with $Z_0^n(0) = n$ and assume $(Z_0^n(t))_{t \geq 0}$ is a sub-critical birth-death process with birth rate r_0 , death rate d_0 , and net growth rate $\lambda_0 = r_0 - d_0 < 0$. We assume that at time t , sensitive cells also give birth to a resistant cell and a sensitive cell at rate $Z_0^n(t) n^{-\alpha}$ for $\alpha \in (0, 1)$ (i.e., a mutation occurs). Each of these birth events results in the creation of a distinct clone of resistant cells but with same fitness. The growth of each clone is modeled as a super-critical birth-death process with birth rate r_1 , death rate d_1 , and net growth rate $\lambda_1 = r_1 - d_1 > 0$. We denote the population of all resistant clones at time t by $(Z_1^n(t))_{t \geq 0}$ with $Z_1^n(0) = 0$.

The ability of resistant cells to proliferate under the treatment allows cancer cells to escape extinction, resulting in cancer recurrence. We define the cancer recurrence time as

$$\gamma_n = \inf\{t \geq 0 : Z_1^n(t) \geq \beta n\}, \text{ with } \beta > 0. \quad (2.1)$$

In this paper, we adopt a simplified approach by setting $\beta = 1$, thereby defining the cancer recurrence time as the first time at which the population of resistant cells reaches the initial population size of sensitive cells. We note that, in practice, the diagnosis of cancer recurrence might not coincide with the moment when the population of resistant cells reaches the precise initial tumor burden. However, our simplification does not substantially affect our subsequent analysis (see Hanagal et al. 2022 for more details).

Remark 2.2 An alternative definition of γ_n is

$$\gamma_n = \inf\{t > 0 : Z_0^n(t) + Z_1^n(t) \geq n\},$$

which represents the first time the total number of cancer cells exceeds the initial tumor size. However, this definition could lead to a very small value for γ_n because of the randomness of the underlying process and the fact that the initial population size is already n . Nevertheless, if we ignore the initial fluctuations in population size and

focus on the cancer recurrence driven by the growth of resistant cells, the difference between the two definitions becomes negligible, as resistant cells will dominate the population with high probability at the time of cancer recurrence.

The model consists of 5 unknown parameters which we denote by $\theta = (\alpha, r_0, d_0, r_1, d_1)$. The objective of this paper is to develop estimators for the aforementioned parameters, using observable data during the cancer recurrence process.

2.2 Observable data during the cancer recurrence process

In Sect. 2.1, we have defined the cancer recurrence time γ_n , which we assume to be an observable quantity clinically. Subsequently, we will present the definitions of several additional important quantities.

Number of Mutant Clones Based on the infinite sites model, we assume each cell that has acquired resistance through mutation in the event of cell birth of sensitive cells gives rise to a distinct clone. Over time, some clones may die out while others continue to proliferate. For ease of exposition, we use “mutant clones” to denote these clones of resistant cells and use $I_n(t)$ to denote the number of mutant clones generated in the time period $(0, t)$ that has an infinite lineage (i.e., clones that do not go extinct).

Remark 2.3 To put it precisely, in a clinical or laboratory context, what we can observe is the number of mutant clones generated in the time period $(0, t)$ and extant at time t , which we denote by $\hat{I}_n(t)$. We opt for the current definition because it is easier to analyze and the difference between the two quantities ($I_n(t)$ and $\hat{I}_n(t)$) is negligible. In Sect. 3.1, we will make this statement precise and subsequently provide its proof.

Remark 2.4 In practice, identifying mutant clones requires prior knowledge of gene lists that, when affected by point mutations or copy number variations, result in drug resistance. Such prior knowledge is available for many cancer therapies. For example, the T315I mutation impedes the efficacy of second-generation tyrosine kinase inhibitors (TKIs) like nilotinib (AMN107) and dasatinib (BMS-354825) (Nussinov et al. 2021), while the T790M and C797S mutations confer resistance to erlotinib and Tagrisso, respectively (Wang et al. 2016).

Simpson’s Index of Mutant Clones Simpson’s Index measures the diversity in size among mutant clones. Specifically, it represents the probability that two resistant cells chosen at random come from the same clone. If we use $X_{i,n}(t)$ to denote the number of resistant cells in the i_{th} clone at time t , then the Simpson’s Index of mutant clones at time t , denoted by $R_n(t)$, is defined as

$$R_n(t) = \sum_{i=1}^{\hat{I}_n(t)} \left(\frac{X_{i,n}(t)}{Z_1^n(t)} \right)^2. \quad (2.2)$$

For completeness, when $Z_1^n(t) = 0$, we define $R_n(t) = 0$.

Remark 2.5 Some of these metrics, such as Simpson's Index, might be observed imprecisely due to potential low-quality data using current experimental techniques and technology. Nonetheless, we expect that with technological advancements, it will be possible to measure the quantities more accurately. In Sect. 4.2, we demonstrate through simulations that imperfect data does not significantly impact the performance of our estimation..

Remark 2.6 Define $\omega_n = \{\omega | \gamma_n(\omega) = \infty\}$, i.e., the event that the tumor never recurs. Note that when ω_n occurs, the observable quantities discussed in this section are ill-defined. However, it does not affect our results as

$$\lim_{n \rightarrow \infty} \mathbb{P}(\gamma_n = \infty) = 0.$$

To show this, we first observe that

$$\mathbb{P}(\gamma_n = \infty) = \mathbb{P}(Z_1^n(t) < n \text{ for any } t > 0),$$

and the desired result follows directly from Corollary 3, which states that

$$\lim_{n \rightarrow \infty} \mathbb{P}(Z_1^n(k\zeta_n) < n) = 0$$

for $k > 1$.

2.3 Previous results

Let $\Phi_i^n(t) = \mathbb{E}[Z_i^n(t)]$, for $i = 0, 1$. We have (see for example Athreya and Ney 1972)

$$\begin{aligned} \Phi_0^n(t) &= ne^{\lambda_0 t}, \\ \Phi_1^n(t) &= \frac{1}{\lambda_1 - \lambda_0} n^{1-\alpha} e^{\lambda_1 t} \left(1 - e^{(\lambda_0 - \lambda_1)t}\right). \end{aligned} \quad (2.3)$$

It is not hard to find that $\Phi_1^n(t)$ is strictly increasing with t . Therefore there is a unique solution ζ_n for the equation

$$\Phi_1^n(t) = n,$$

and we can obtain that $\zeta_n \sim \frac{\alpha}{\lambda_1} \log n$. From our previous work (Foo and Leder 2013; Leder and Wang 2021), we have

$$\lim_{n \rightarrow \infty} \mathbb{P}\left(\left|\gamma_n - \frac{\alpha}{\lambda_1} \log n\right| > \epsilon\right) = 0, \quad (2.4)$$

$$\lim_{n \rightarrow \infty} \frac{1}{n^{1-\alpha}} \mathbb{E}[I_n(\zeta_n)] = -\frac{\lambda_1}{\lambda_0 r_1}, \text{ and} \quad (2.5)$$

$$\lim_{n \rightarrow \infty} n^{1-\alpha} \mathbb{E}[R_n(\zeta_n)] = \frac{2r_1(\lambda_1 - \lambda_0)^2}{\lambda_1(2\lambda_1 - \lambda_0)}. \quad (2.6)$$

To be specific, we have

$$\mathbb{E}[I_n(\zeta_n)] = -\frac{\lambda_1}{\lambda_0 r_1} n^{1-\alpha} (1 - e^{\lambda_0 \zeta_n}). \quad (2.7)$$

In Eq. (2.4), γ_n is an clinically observable quantity, whereas α and λ_1 are parameters in our model that we hope to estimate. This establishes a link between the model's ‘unobservable’ parameters and an ‘observable’ quantity. Our goal is to further explore connections like this to develop consistent estimators for our model parameters. Equations (2.5) and (2.6) are two other results that connect observable quantities (the number of mutant clones and Simpson’s Index) with model parameters. However, these results have limited utility for constructing estimators in our context for two main reasons: Firstly, ζ_n is a deterministic approximation for γ_n and it is not observable in practice. Secondly, estimators constructed based on the convergence of expectation require a large dataset (see Leder and Wang 2021), which is not only costly but also does not ensure the consistency of the estimators. Furthermore, expectation based estimators do not allow for patient specific parameters and would instead estimate a mean parameter value for the population. Consequently, we will extend these results [(2.5) and (2.6)] to the random time γ_n and establish the corresponding convergence in probability results.

3 Theoretical results

In this section, we first present our main results on the convergence of ‘observable’ data during the cancer recurrence processes (Sect. 3.1). Subsequently, leveraging these findings, we develop consistent estimators for the parameters of our model (Sect. 3.2).

3.1 Convergence results of observable data

We first establish a stronger convergence result for the cancer recurrence time γ_n . In Foo and Leder (2013), the authors showed that $\gamma_n - \zeta_n$ converges to 0 in probability. In Theorem 1, we show that this result can be strengthened by specifying the convergence rate.

Theorem 1 (Convergence rate of γ_n) *For any $\epsilon > 0$, $u < (1 - \alpha)/2$, we have*

$$\lim_{n \rightarrow \infty} \mathbb{P}(n^u |\gamma_n - \zeta_n| > \epsilon) = 0. \quad (3.1)$$

Proof See Sect. B. □

Next, we show a convergence in probability result for the number of mutant clones, $I_n(\gamma_n)$, observed at the time of cancer recurrence.

Theorem 2 (Convergence of $I_n(\gamma_n)$) *For any $\epsilon > 0$, $u < \min\{(1-\alpha)/2, -\lambda_0\alpha/2\lambda_1\}$, we have*

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(n^u \left| \frac{1}{n^{1-\alpha}} I_n(\gamma_n) + \frac{\lambda_1}{\lambda_0 r_1} \right| > \epsilon \right) = 0. \quad (3.2)$$

Proof See Sect. C. □

In the aforementioned Remark 2.3, the term $I_n(t)$ represents the number of clones generated before time t that do not go extinct. It is important to note that at any given time t , it is impossible to know whether a clone will go extinct or not. However, it can be demonstrated that the actual observation $\hat{I}_n(t)$, defined as the number of clones that emerge prior to time t and remain extant at time t , has a negligible difference with $I_n(t)$.

Proposition 1 *There exists $c_1 > 0$, such that for any $t > 0$*

$$\frac{1}{n^{1-\alpha}} \mathbb{E} [\hat{I}_n(t) - I_n(t)] \leq \frac{c_1 d_1}{r_1(\lambda_0 + \lambda_1)} (e^{\lambda_0 t} - e^{-\lambda_1 t})$$

if $\lambda_0 + \lambda_1 \neq 0$; and

$$\frac{1}{n^{1-\alpha}} \mathbb{E} [\hat{I}_n(t) - I_n(t)] \leq \frac{c_1 d_1}{r_1} t e^{-\lambda_1 t}$$

otherwise.

Proof We use $A(t)$ to denote the arrival process of mutant clones that are generated within the interval $(0, t)$ and go extinct eventually, but survive until time t . $A(t)$ is clearly a conditional non-homogeneous Poisson process with mean

$$\begin{aligned} \mathbb{E}[A(t)] &= \mathbb{E}[\mathbb{E}[A(t)|Z_0(s), s \leq t]] \\ &= \mathbb{E} \left[\int_0^t Z_0(s) n^{-\alpha} \mathbb{P}(t-s < \tau_0 < \infty) ds \right] \\ &= \mathbb{P}(\tau_0 < \infty) \mathbb{E} \left[\int_0^t Z_0(s) n^{-\alpha} \mathbb{P}(t-s < \tau_0 | \tau_0 < \infty) ds \right] \\ &\stackrel{(a)}{=} \mathbb{P}(\tau_0 < \infty) n^{1-\alpha} \int_0^t e^{\lambda_0 s} \mathbb{P}(\tau_0 > t-s | \tau_0 < \infty) ds, \end{aligned}$$

where τ_0 represents the extinction time of a mutant clone starting from a single resistant cell, and we use Fubini's theorem in equality (a). It is established in the literature (see the proof of Durrett et al. 2016, Lemma 2) when conditioning on the event of eventual extinction, i.e. $\tau_0 < \infty$, a supercritical birth-death process becomes a subcritical process with birth and death rates interchanged. It implies that

$$\mathbb{P}(\tau_0 > t-s | \tau_0 < \infty) = \mathbb{P}(\tilde{\tau}_0 > t-s),$$

where $\tilde{\tau}_0$ represents the extinction time for a subcritical birth-death process starting from a single cell with birth rate d_1 and death rate r_1 . From Proposition 1 of Jagers et al. (2007), for some $c_1 > 0$, the tail probability of the extinction time for the aforementioned subcritical birth-death process is bounded by

$$\mathbb{P}(\tilde{\tau}_0 > t) \leq c_1 e^{-\lambda_1 t}.$$

Therefore, if $\lambda_0 + \lambda_1 \neq 0$, we have

$$\begin{aligned} \mathbb{E}[A(t)] &= \mathbb{P}(\tau_0 < \infty) n^{1-\alpha} \int_0^t e^{\lambda_0 s} \mathbb{P}(\tau_0 > t-s | \tau_0 < \infty) ds \\ &\leq \mathbb{P}(\tau_0 < \infty) n^{1-\alpha} e^{-\lambda_1 t} \int_0^t c_1 e^{(\lambda_0 + \lambda_1)s} ds \\ &= \frac{c_1 d_1}{r_1(\lambda_0 + \lambda_1)} n^{1-\alpha} (e^{\lambda_0 t} - e^{-\lambda_1 t}), \end{aligned}$$

which implies that

$$\frac{1}{n^{1-\alpha}} \mathbb{E}[\hat{I}_n(t) - I_n(t)] = \frac{1}{n^{1-\alpha}} \mathbb{E}[A(t)] \leq \frac{c_1 d_1}{r_1(\lambda_0 + \lambda_1)} (e^{\lambda_0 t} - e^{-\lambda_1 t}).$$

The case where $\lambda_0 + \lambda_1 = 0$ can be analyzed in a similar way and thus we omit the details here. $\square$

Utilizing Proposition 1 and Theorem 2, we demonstrate that the actual observable number of clones also converges in probability.

Corollary 1 For any $\epsilon > 0$, $u < \min\{(1-\alpha)/2, -\lambda_0\alpha/2\lambda_1\}$, we have

$$\lim_{n \rightarrow \infty} \mathbb{P}\left(n^u \left| \frac{1}{n^{1-\alpha}} \hat{I}_n(\gamma_n) + \frac{\lambda_1}{\lambda_0 r_1} \right| > \epsilon\right) = 0. \quad (3.3)$$

Proof By Theorem 2, it suffices to show

$$\lim_{n \rightarrow \infty} \mathbb{P}\left(n^u \left(\frac{1}{n^{1-\alpha}} \hat{I}_n(\gamma_n) - \frac{1}{n^{1-\alpha}} I_n(\gamma_n) \right) > \epsilon\right) = 0.$$

Because $\hat{I}_n(\gamma_n) \geq I_n(\gamma_n)$, we can apply the Markov Inequality and obtain that

$$\begin{aligned} &\mathbb{P}\left(n^u \left(\frac{1}{n^{1-\alpha}} \hat{I}_n(\gamma_n) - \frac{1}{n^{1-\alpha}} I_n(\gamma_n) \right) > \epsilon\right) \\ &\leq \mathbb{P}\left(n^u \left(\frac{1}{n^{1-\alpha}} \hat{I}_n(\gamma_n) - \frac{1}{n^{1-\alpha}} I_n(\gamma_n) \right) > \epsilon, |\gamma_n - \zeta_n| < \epsilon\right) + \mathbb{P}(|\gamma_n - \zeta_n| > \epsilon) \\ &= n^{u-1+\alpha} \mathbb{E}\left[\hat{I}_n(\gamma_n) - I_n(\gamma_n); |\gamma_n - \zeta_n| < \epsilon\right] / \epsilon + \mathbb{P}(|\gamma_n - \zeta_n| > \epsilon). \end{aligned}$$

By Theorem 1, the second term converges to 0. Meanwhile, we have

$$\begin{aligned} & n^{u-1+\alpha} \mathbb{E} \left[\hat{I}_n(\gamma_n) - I_n(\gamma_n); |\gamma_n - \zeta_n| < \epsilon \right] \\ & \leq n^{u-1+\alpha} \mathbb{E} \left[\hat{I}_n(\gamma_n) - I_n(\zeta_n - \epsilon); |\gamma_n - \zeta_n| < \epsilon \right] \\ & = n^{u-1+\alpha} \mathbb{E} \left[\hat{I}_n(\zeta_n - \epsilon) - I_n(\zeta_n - \epsilon); |\gamma_n - \zeta_n| < \epsilon \right] \end{aligned} \quad (3.4)$$

$$+ n^{u-1+\alpha} \mathbb{E} \left[\hat{I}_n(\gamma_n) - \hat{I}_n(\zeta_n - \epsilon); |\gamma_n - \zeta_n| < \epsilon \right]. \quad (3.5)$$

By Proposition 1, when $u < \min\{(1 - \alpha)/2, -\lambda_0\alpha/2\lambda_1\}$, we have (3.4) $\rightarrow 0$. By a similar argument to that in the proof of Lemma 3 (see Sect. C), we have (3.5) $\rightarrow 0$, which gives us

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(n^u \left(\frac{1}{n^{1-\alpha}} \hat{I}_n(\gamma_n) - \frac{1}{n^{1-\alpha}} I_n(\gamma_n) \right) > \epsilon \right) = 0.$$

□

Next, we present a convergence in probability result for the Simpson's Index in Theorem 3. We begin with the following proposition which shows that with high probability, resistant cells survive to the deterministic approximation of cancer recurrence time ζ_n , which paves the way for our later analysis for Simpson's Index. In particular define the set $\rho_n = \{\omega | Z_1^n(\zeta_n) > 0\}$ then we have the following result.

Proposition 2 As $n \rightarrow \infty$, $\mathbb{P}(\rho_n) \rightarrow 1$.

Proof From (A.2), we have

$$\begin{aligned} \mathbb{P}(\rho_n^C) &= \mathbb{P}(Z_1^n(\zeta_n) = 0) \leq \mathbb{P}(|Z_1^n(\zeta_n) - n| \geq n) \\ &\leq \frac{\mathbb{E}[(Z_1^n(\zeta_n) - n)^2]}{n^2} = \Theta(n^{\alpha-1}) \rightarrow 0. \end{aligned}$$

□

Furthermore, we have the following convergence in probability result for the Simpson's Index.

Theorem 3 (Convergence of $R_n(\gamma_n)$) For any $\epsilon > 0$, $\alpha < 1$, $u < \min\{(1 - \alpha)/4, (\lambda_1 - \lambda_0)\alpha/2\lambda_1\}$, we have

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(n^u \left| n^{1-\alpha} R_n(\gamma_n) - \frac{2r_1(\lambda_1 - \lambda_0)^2}{\lambda_1(2\lambda_1 - \lambda_0)} \right| > \epsilon \right) = 0. \quad (3.6)$$

Proof Outline: From Proposition 2 it will suffice to establish

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(n^u \left| n^{1-\alpha} R_n(\gamma_n) - \frac{2r_1(\lambda_1 - \lambda_0)^2}{\lambda_1(2\lambda_1 - \lambda_0)} \right| > \epsilon, \rho_n \right) = 0.$$

To prove this result, we initially demonstrate the convergence of the Simpson Index at the deterministic time ζ_n , as detailed in Lemma 5. In Lemma 6 we establish the proximity of $R_n(\gamma_n)$ to $R_n(\zeta_n)$. The detailed proofs can be found in Sect. D.

Theorems 1, 2 and 3 ensure that, with high probability, the cancer recurrence time γ_n , the number of mutant clones $I_n(\gamma_n)$, and the Simpson's Index of mutant clones $R_n(\gamma_n)$ observed from a single patient sample are very close to their deterministic limits, respectively. In Sect. 3.2, we use these results to develop a set of consistent estimators for our model parameters.

3.2 Consistent estimators for model parameters

In this section, we develop a set of consistent estimators for our model parameters, including the growth rate of sensitive cells λ_0 , the growth rate of resistant cells λ_1 , the birth rate of resistant cells r_1 , and the mutation rate related parameter α . These parameters are fundamental to the dynamics of cancer recurrence. The development of consistent estimators for these parameters can help practitioners gain a deeper understanding of the dynamics of cancer recurrence and for facilitating the decision-making process regarding patient treatment plans. For ease of notation, we use $Z_0^n \equiv Z_0^n(\gamma_n)$, $I_n \equiv I_n(\gamma_n)$ and $R_n \equiv R_n(\gamma_n)$.

Theorem 4 For $\alpha < -\frac{\lambda_1}{\lambda_0}$, define

$$\begin{aligned}\hat{\lambda}_0^{(n)} &= \frac{1}{\gamma_n} \log \frac{Z_0^n}{n}, \\ \hat{\lambda}_1^{(n)} &= -\frac{\hat{\lambda}_0^{(n)}}{U_n}, \\ \hat{r}_1^{(n)} &= \left(\frac{1}{U_n} + 1 \right) \frac{n \hat{\lambda}_1^{(n)}}{I_n e^{\hat{\lambda}_1^{(n)} \gamma_n}}, \text{ and} \\ \hat{\alpha}^{(n)} &= 1 - \log_n I_n + \log_n \left(\frac{\hat{\lambda}_1^{(n)}}{-\hat{\lambda}_0^{(n)} \hat{r}_1^{(n)}} \right),\end{aligned}$$

where $U_n = \frac{\sqrt{I_n R_n}}{\sqrt{I_n R_n} - 2} - 1$. Then, $\hat{\lambda}_0^{(n)}$, $\hat{\lambda}_1^{(n)}$, $\hat{r}_1^{(n)}$ and $\hat{\alpha}^{(n)}$ are consistent estimators of λ_0 , λ_1 , r_1 and α respectively.

Proof See Sect. E. □

The implication of the assumption on α is that if $\alpha \geq -\frac{\lambda_1}{\lambda_0}$, there is a non-negligible probability that sensitive cells go extinct at the time of cancer recurrence. This is important because our estimator for λ_0 relies on Z_0^n , the number of sensitive cells at the time of recurrence. Substantial clinical and experimental evidence indicates that for a wide range of cancer types and treatments, sensitive cells do not go extinct upon cancer recurrence. For example, studies on non-small cell lung cancer with T790M mutation-induced resistance to tyrosine kinase inhibitors (TKI) such as erlotinib have

shown that mutation-free sensitive cells persist at progression when treated with second and third generation EGFR inhibitors (Chaft et al. 2011; Ichihara and Lovly 2015; Piotrowska et al. 2015). The persistence of sensitive cells is also common under adaptive therapies, which are designed to maintain a population of sensitive cells to suppress the proliferation of resistant cells, as discussed in Strobl et al. (2022), Gatenby et al. (2009).

Theorem 4 establishes the consistency of our estimators, ensuring their effectiveness when the initial size of the tumor is sufficiently large. Moreover, for $\hat{\lambda}_0^{(n)}$ and $\hat{\lambda}_1^{(n)}$, a lower bound can be established for the rate of convergence for both estimators. In particular, in the proof of Theorem 4, we obtain that

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(n^u \left| \hat{\lambda}_0^{(n)} - \lambda_0 \right| > \epsilon \right) = 0 \quad \text{for } u < \min\{(1 - \alpha)/2, 1/2 + \alpha\lambda_0/2\lambda_1\}, \text{ and}$$

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(n^u \left| \hat{\lambda}_1^{(n)} - \lambda_1 \right| > \epsilon \right) = 0 \quad \text{for } u < \min\{(1 - \alpha)/4, -\lambda_0\alpha/2\lambda_1, 1/2 + \alpha\lambda_0/2\lambda_1\}.$$

To the best of our knowledge, we are among the first in presenting an analysis of the convergence rate for estimators of key parameters in tumor dynamics. As will be discussed in Sect. 4, this analysis leads to an accurate estimation of these parameters within practical and realistic settings.

Remark 3.1 It should be noted that the formulation of our estimators utilizes data on the number of mutant clones and their Simpson's Index at the recurrence time. This data yields the same information as the mutant clone size's empirical mean and second moment. Specifically, while it is possible to devise estimators based on the first and second moments of mutant clone size, our preference is to employ the number of clones and Simpson's Index, which offers a more nuanced understanding of heterogeneity.

4 Simulation results and extensions

In this section, we explore the performance of our estimators through numerical simulations.

4.1 Convergence of estimators

Theoretically, our estimators are posited to work with a sample of recurrent tumor from a patient, provided several conditions are met. These include the survival of sensitive cells until the time of cancer recurrence, a sufficiently large tumor size, and a constraint on the model parameters ($0 < \alpha < \min\{-\lambda_1/\lambda_0, 1\}$). In this section, we present simulation results and a detailed discussion aimed at identifying the threshold for tumor size that ensures a good performance of our estimators.

We consider an example system with $\alpha = 0.5$, $\lambda_0 = -0.5$, $r_1 = 1.5$, and $d_1 = 1$. Note that $\lambda_1 = r_1 - d_1 = 0.5$. Recall that we denote by n the initial tumor burden (i.e., the initial population size of sensitive cells). We measure the performance of our estimators for $n = 100 \times 2^i$, where $i \in \{0, 1, \dots, 20\}$. Specifically, for each value of n , we simulate the corresponding system until the population of resistant cells reaches

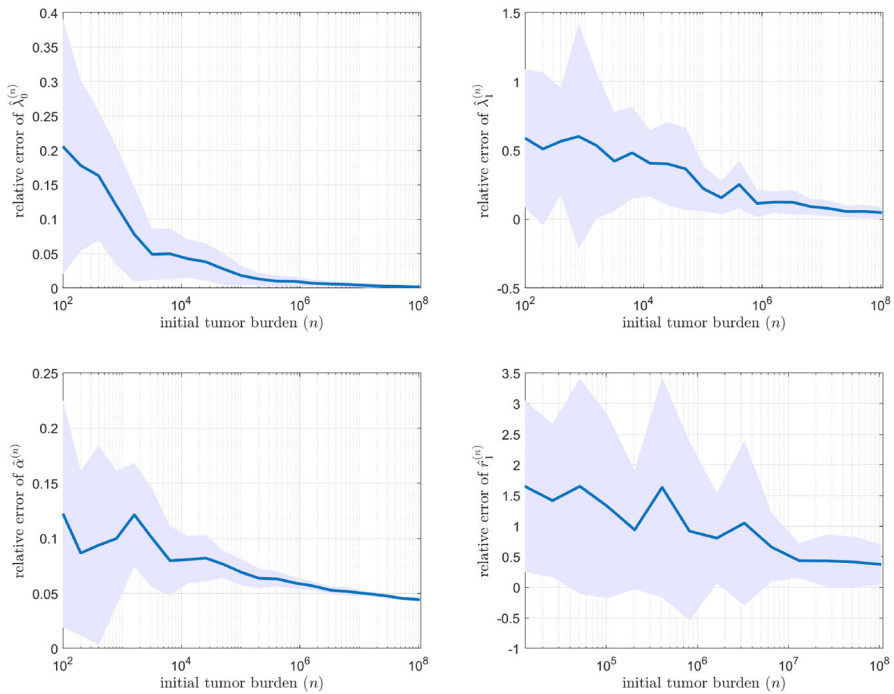


Fig. 1 In this experiment, we set $\alpha = 0.5$, $\lambda_0 = -0.5$, $r_1 = 1.5$, $d_1 = 1$ and $\lambda_1 = 0.5$. We let n vary from 100 to 100×2^{20} . For each value of n , we repeat the experiment $k = 20$ times. We represent the mean relative errors of each estimator with a dark blue line and illustrate the standard deviation using a shaded area (color figure online)

the initial tumor burden n . We then record the recurrence time (γ_n), the population size of sensitive cells at cancer recurrence ($Z_0(\gamma_n)$), the number of mutant clones ($I_n(\gamma_n)$), and Simpson's Index ($R_n(\gamma_n)$). We can compute the estimators $\hat{\lambda}_0^{(n)}$, $\hat{\lambda}_1^{(n)}$, $\hat{r}_1^{(n)}$ and $\hat{\alpha}_1^{(n)}$ by Theorem 4, and then calculate the relative errors of these estimators, i.e.,

$$\lambda_0^{error} = \frac{|\hat{\lambda}_0^{(n)} - \lambda_0|}{-\lambda_0}, \quad \lambda_1^{error} = \frac{|\hat{\lambda}_1^{(n)} - \lambda_1|}{\lambda_1}, \quad \alpha^{error} = \frac{|\hat{\alpha}^{(n)} - \alpha|}{\alpha}, \quad r_1^{error} = \frac{|\hat{r}_1^{(n)} - r_1|}{r_1}.$$

For each value of n , we repeat such a process for $k = 20$ times and report the mean value and standard deviation of λ_0^{error} , λ_1^{error} , α^{error} and r_1^{error} in Fig. 1.

Observations drawn from Fig. 1 reveal that the mean values of relative errors associated with all estimators exhibit a decreasing trend towards 0 as n increases. Concurrently, the standard deviations, represented by the shaded areas in the figure, similarly diminish to 0 with the increase in n . These observations align with the theoretical findings presented in Theorem 4, confirming the consistency of our estimators.

Notably, these estimators exhibit dissimilar levels of performance. In particular, $\hat{\lambda}_0^{(n)}$ demonstrates superior performance, an observation that aligns with the theoretical results established in the proof of Theorem 4, which states that $\hat{\lambda}_0^{(n)}$ approaches λ_0 at a convergence rate of at least $n^{-\min\{(1-\alpha)/2, (1+\lambda_0\alpha/\lambda_1)/2\}}$. However, the convergence rate of $\hat{\lambda}_1^{(n)}$ is only ensured to be at most $n^{-\min\{(1-\alpha)/4, -\lambda_0\alpha/2\lambda_1, (1+\lambda_0\alpha/\lambda_1)/2\}}$. On the other hand, there are no assurances about the convergence rates of $\alpha^{(n)}$ and $\hat{r}_1^{(n)}$. Additionally, we note that the rate at which $\hat{r}_1^{(n)}$ converges is the slowest. The convergence of $\hat{r}_1^{(n)}$ depends on the convergence of the term $e^{(\lambda_1 - \hat{\lambda}_1^{(n)})\zeta_n}$ (see the proof of Theorem 4 in Sect. E). However, based on the convergence result of $\hat{\lambda}_1^{(n)}$, the theoretical guarantee for the convergence of $(\lambda_1 - \hat{\lambda}_1^{(n)})\zeta_n$ only holds when ζ_n is less than $n^{\min\{(1-\alpha)/4, -\lambda_0\alpha/2\lambda_1, (1+\lambda_0\alpha/\lambda_1)/2\}}$. Given that ζ_n is approximately $\frac{\alpha}{\lambda_1} \log n$, in order to ensure that the condition is satisfied, the value of n might exceed 10^{11} depending on the setting of parameters. This magnitude surpasses the capabilities of our current simulation framework. However, it is within the realm of possibility for certain types of tumors to achieve such a numerical scale. A 10-cm cancerous tumor is estimated to contain approximately 10^{11} cells (Narod 2012). This size is not uncommon in various tumor types, with instances documented in breast carcinoma (Giuliano et al. 1995) as well as nasopharyngeal carcinoma (Sze et al. 2004), among others.

4.2 Performance of estimators in the presence of undetectable clones

In this study, we have demonstrated, both theoretically and via simulation, the good performance of our estimators of the model parameters λ_0 , λ_1 , α , and r_1 in contexts where there is a large initial population size n of drug sensitive cells. However, the accuracy of our estimators relies heavily on calculating the Simpson Index. This calculation requires detailed information about the sizes of all mutant clones at the time of recurrence. A significant challenge arises due to the limitations of current genomic sequencing technologies, which are incapable of identifying mutant clones of small size. Previous studies (Roth et al. 2014; Stead et al. 2013) indicate that the minimum detectable clone size using current technologies is around 2% of the total tumor size. Consequently, it is necessary to modify the Simpson Index computation in order to account for these undetectable clones, using only the data of detectable clones. To assess the effectiveness of this strategy, we conduct a simulation experiment that takes into account the presence of clones that fall below the detection threshold.

In our simulation experiments, we fix $\alpha = 0.8$, $r_0 = 1.3$, $d_0 = 1.5$, $r_1 = 2.0$, and $d_1 = 1.2$, while progressively increasing the initial tumor burden n . Each experiment consists of three distinct scenarios for evaluating our estimators: a baseline control and two test settings. In test settings, we exclude clone data that are smaller than 2% and 10% of the total tumor size at the time of recurrence, respectively. This approach allows us to assess the performance of our estimators in scenarios with varying degrees of data availability.

As illustrated in Fig. 2, excluding clone size data below 10% of the total tumor population may significantly affect the performance of our estimators. In contrast, the

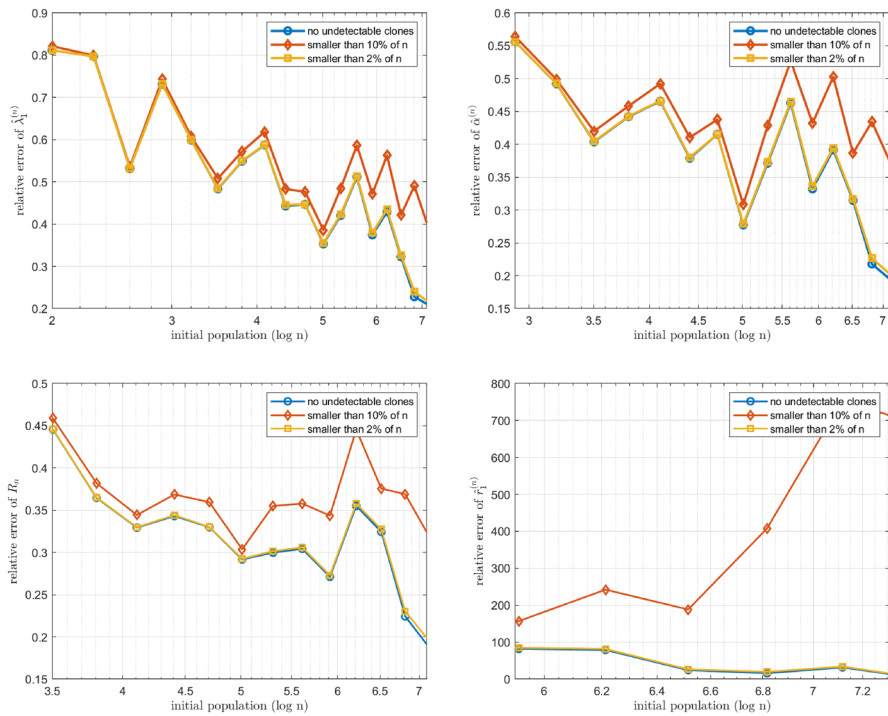


Fig. 2 In this experiment, we conduct a comparative analysis across three experimental setups. We set $\alpha = 0.8$, $\lambda_0 = -0.2$, $\lambda_1 = 0.8$ and $r_1 = 2.0$. The first setup is a baseline scheme without restrictions on clone size observation; the second setup imposes a threshold, only considering clones that constitute at least 2% of the total population; and the third raises this threshold to 10%. We assess the impact of these varying observation restrictions on the performance of the estimators for λ_1 , α and r_1 , as well as the computation for Simpson Index R_n

omission of clones smaller than 2% of the total tumor population appears to have a negligible impact. This observation implies that, given the capabilities of current technologies to detect clones as small as 2% of the total tumor size, the estimators constructed using our method are still reliable.

4.3 Stability analysis

As established in Theorem 4, the performance of our estimators relies on the assumption that $0 < \alpha < \min\{1, -\lambda_1/\lambda_0\}$. In this section, we examine the robustness of our estimators across a range of parameter settings, particularly near the critical boundary conditions, such as when α is close to either $\min\{1, -\lambda_1/\lambda_0\}$ or 0. To facilitate a clear interpretation of the results, we conduct three sets of numerical experiments. For each set, we modify a single parameter while keeping the others constant, enabling us to isolate the impact of each parameter.

In the first set of experiments, we fix $r_1 = 1.5$, $d_1 = 1.0$, and $\alpha = 0.5$. The values for r_0 and d_0 are uniformly sampled from (0.8, 1.2) and (1.3, 1.7) respectively,

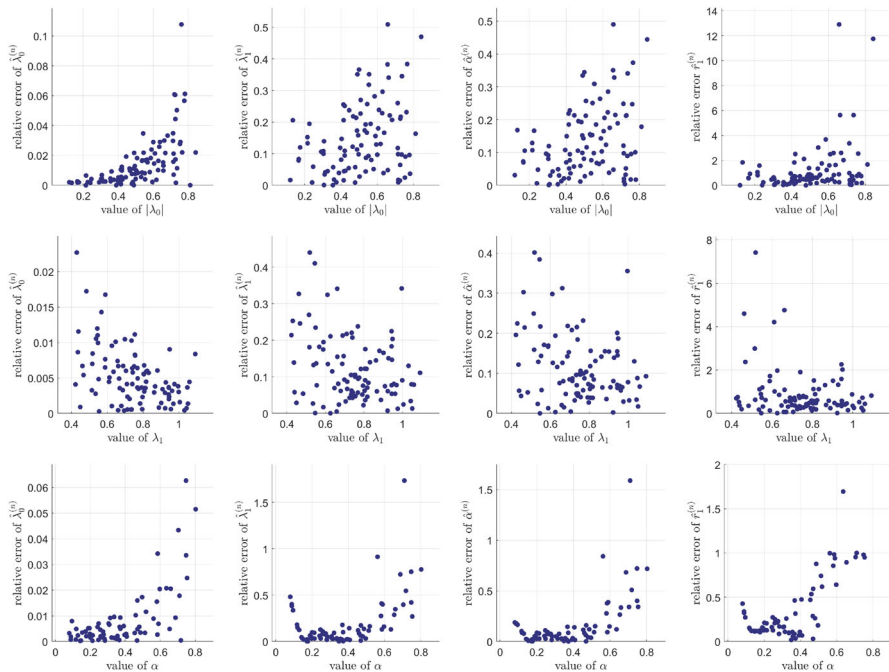


Fig. 3 In this experiment, we vary the model parameters and check the stability of our estimators. In the first setup, We fix $r_1 = 1.5$, $d_1 = 1.0$, and $\alpha = 0.5$, and select λ_0 uniformly from the interval $(-0.9, -0.1)$ (see the first row of Fig. 3). In the second setup, we fix $r_0 = 1.0$, $d_0 = 1.5$, and $\alpha = 0.5$ and select λ_1 uniformly from the interval $(0.3, 0.9)$ (see the second row of Fig. 3). Lastly, we set $r_0 = d_0 = 1.0$ and $r_1 = d_1 = 1.5$, and select α uniformly from the interval $(0, 1)$ (see the third row of Fig. 3). Under all setups, we fix the initial tumor burden to be $n = 1 \times 10^6$, and conduct numerical simulations to estimate λ_0 , λ_1 , α , and r_1 . The resulting relative errors for these estimations, corresponding to each of the defined setups, are presented in Fig. 3

ensuring that λ_0 falls within $(-0.9, -0.1)$. In this case, we have $-\lambda_0 < \lambda_1/\alpha = 1$. In the second set of experiments, we fix $r_0 = 1.0$, $d_0 = 1.5$, and $\alpha = 0.5$. The values for r_1 and d_1 are uniformly sampled from $(1.4, 1.8)$ and $(0.7, 1.1)$ respectively, ensuring that λ_1 falls within $(0.3, 1.1)$. In this case, we have $\lambda_1 > -\alpha\lambda_0 = 0.25$. In the third set of experiments, we fix $r_0 = 1$, $d_0 = 1$, $r_1 = 1.5$, and $d_1 = 1.5$ respectively. The value of α is uniformly sampled from $(0, 1)$.

The numerical results depicted in Fig. 3 demonstrate that our estimators exhibit better performance in certain scenarios compared to others. These include scenarios where the absolute value of the growth rate of sensitive cells, $|\lambda_0|$, is low; the growth rate of resistant cells, λ_1 , is high; or the mutation rate, α , is moderately small. A mathematical interpretation of this result can be linked to the boundary condition. By defining the parameter gap as $\Delta = 1 - \alpha|\lambda_0|/\lambda_1$, it becomes evident that a decrease in $|\lambda_0|$ or an increase in λ_1 , or a decrease in α , results in a reduction of the parameter gap Δ . Note that

$$Z_0^n(\gamma_n) \approx \Phi_0^n(\zeta_n) \sim n^\Delta.$$

Hence, the population size of sensitive cells at cancer recurrence tends to be small when Δ is small, which results in a high relative variability of $Z_0^n(\gamma_n)$. The increased relative variability of $Z_0^n(\gamma_n)$ then affects the estimation of other parameters. We also note that as α approaches 0, which indicates a high mutation rate, the estimators for λ_1 , α , and r_1 exhibit notably poor performance. This issue can be attributed to the emergence of a large number mutant clones, resulting in a very small Simpson's Index which increases the relative error on the estimation of R_n .

Performance improvement via bootstrapping

Based on the stability analysis, the performance of our estimators declines when the parameter α approaches 1. This presents a practical challenge in implementing these estimators, especially for tumor types with low mutation rates.

To address this issue, we propose a bootstrapping technique to enhance the accuracy in calculating the Simpson Index. In particular, we calculate Simpson's Index using bootstrap samples which include the largest 20% of clones, and the randomly sampled 5/8th of the remaining clones. After generating 1000 bootstrap samples we average the resulting Simpson's Index values to form an improved estimator. We examined the setting with parameters $n = 10^7$, $\alpha = 0.8$, $\lambda_0 = -0.2$, and $\lambda_1 = 0.8$. We conducted a comparative analysis to assess the effectiveness of the bootstrapping technique in improving estimator accuracy. By repeating the estimation process 10 times and plotting the relative errors for both the estimators and the Simpson Index, we were able to compare the results before and after applying bootstrapping. To be specific, we denote the Simpson Index as well as estimators after bootstrapping as $\tilde{R}_n$, $\tilde{\lambda}_1^{(n)}$, $\tilde{r}_1^{(n)}$, and $\tilde{\alpha}^{(n)}$. We calculate

$$\left| \frac{\tilde{R}_n - \mathbb{E}[R_n]}{R_n - \mathbb{E}[R_n]} \right|, \left| \frac{\tilde{\lambda}_1^{(n)} - \lambda_1}{\hat{\lambda}_1^{(n)} - \lambda_1} \right|, \left| \frac{\tilde{\alpha}^{(n)} - \alpha}{\hat{\alpha}^{(n)} - \alpha} \right| \text{ and } \left| \frac{\tilde{r}_1^{(n)} - r_1}{\hat{r}_1^{(n)} - r_1} \right|$$

where we use $n^{\alpha-1} \frac{2r_1(\lambda_1 - \lambda_0)^2}{\lambda_1(2\lambda_1 - \lambda_0)}$ to approximate $\mathbb{E}[R_n]$ according to (2.6). As illustrated in Fig. 4, the implementation of bootstrapping techniques improves the precision in calculating the Simpson Index and the associated estimators. This enhancement is particularly notable in the case of r_1 , which typically presents a significant challenge in the estimation without bootstrapping (Table 1).

4.4 Robustness analysis on carrying capacity

It is important to note that our model implicitly assumes an unbounded capacity for cancer cells, which is reflected through the consistent growth rate λ_1 of resistant cells. This assumption, while facilitating the analysis, might not fully capture the complexities that arise in situations where the growth of recurrent tumors is restricted by factors such as supply of nutrients or spatial constraints. Therefore, in this section, we assess the robustness of our estimators in a more realistic scenario that takes into account a maximum capacity for cancer cells. Specifically, we propose a modified

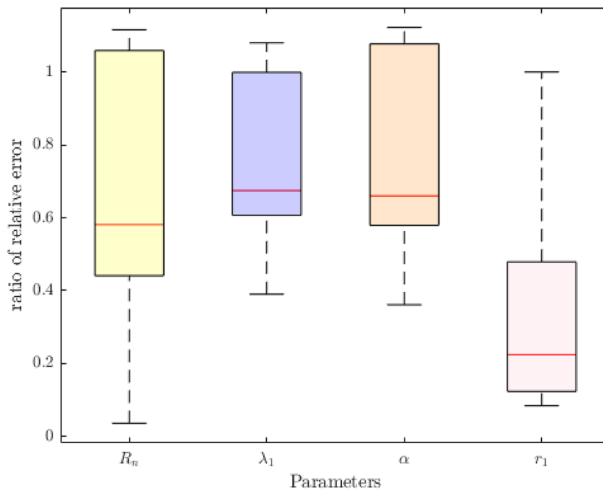


Fig. 4 We set $n = 10^7$, $\alpha = 0.8$, $\lambda_0 = -0.2$, and $\lambda_1 = 0.8$, and then generate 10 simulated data sets. For each data set we generate estimators using Theorem 4. We then employ bootstrapping techniques to refine these estimates. For a visual comparison of the bootstrapping technique's effectiveness, we generated box plots illustrating the distribution of the ratio of relative errors post-bootstrapping to those pre-bootstrapping. The median of these distributions is highlighted by a red line (color figure online)

Table 1 Pseudocode for improved parameter estimation via bootstrapping

Output Simpson Index and estimators after bootstrapping $\tilde{R}_n$, $\tilde{\lambda}_1^{(n)}$, $\tilde{r}_1^{(n)}$, and $\tilde{\alpha}^{(n)}$

1. Initialize parameters: $\lambda_0, \lambda_1, \alpha, r_1, M$.
2. Run simulation experiment, and generate observations Z_0^n, I_n , as well as clone list $\{X_{i,n}\}_{i \geq 1}$
3. For $k = 1 : M$
 - Generate random subset of $\{X_{i,n}\}_{i \geq 1}$, denoted by $\{X_{i,n}^k\}_{i \geq 1}$
 - Order clone sizes and denote them to be

$$X_{(1),n} > X_{(2),n} > \dots > X_{(I_n),n}$$
 - Set $a = \lfloor I_n/5 \rfloor$ and $b = \lfloor 7I_n/10 \rfloor$
 - Let $(X_{1,n}^k, \dots, X_{a,n}^k) = (X_{(1),n}, \dots, X_{(a),n})$ which we denote as set T .
 - For $i \in [a+1, b]$, choose $X_{i,n}^k$ at random from $\{X_{i,n}\}_{i \geq 1} \setminus T$
 - Calculate $\tilde{R}_n^{(k)}$ using $\{X_{i,n}^k\}_{i \geq 1}$ based on (2.2).
4. $\tilde{R}_n$ = average of $\tilde{R}_n^{(k)}$. Derive $\tilde{\lambda}_1^{(n)}, \tilde{r}_1^{(n)}, \tilde{\alpha}^{(n)}$ using $Z_0^n, I_n, \tilde{R}_n$ according to Theorem 4

model where the birth rate of resistant cells is dynamically adjusted based on the total cancer cell population. Mathematically, with C representing the maximum capacity for cancer cells, the adjusted birth rate of resistant cells is:

$$r_1(t) = \max \left(0, \lambda_1 \left(1 - \frac{Z_0^n(t) + Z_1^n(t)}{C} \right) \right) + d_1.$$

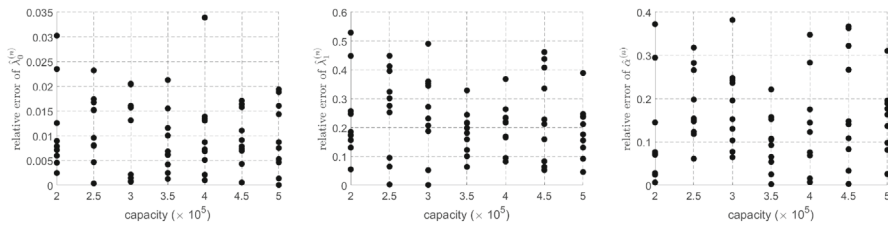


Fig. 5 In this experiment, we set $n = 10^5$, $\alpha = 0.5$, $\lambda_0 = -0.5$, and $\lambda_1 = 0.5$. We vary the capacity C from $2n$ to $5n$ and record the relative error of the estimators across different capacity settings

We numerically investigate this modified model under the following parameter settings: $n = 10^5$, $\alpha = 0.5$, $\lambda_0 = -0.5$, and $\lambda_1 = 0.5$. We vary the capacity parameter C from 2×10^5 to 5×10^5 and examine how our estimators behave and perform under these modified conditions.

As illustrated in Fig. 5, the carrying capacity appears to have a negligible impact on the performance of our estimators for λ_0 , λ_1 and α . To quantify these observations, we conducted a linear regression analysis of carrying capacity versus relative error for the three estimators $\hat{\lambda}_0^{(n)}$, $\hat{\lambda}_1^{(n)}$ and $\hat{\alpha}^{(n)}$. The 90% confidence intervals for the slope parameters of these regressions are $[-0.00131, 0.00010]$, $[-0.0036, 0.0070]$ and $[-0.026, 0.033]$, respectively. These results indicate that changes in carrying capacity do not significantly influence any of the estimators.

5 Summary and discussion

In summary, this study leverages a two-type birth and death model for cancer recurrence, where the cancer cells are divided into sensitive cells and resistant cells according to their fitness under pharmacological treatments. We examined several quantities that can be observed during the cancer recurrence process, namely, recurrence time, number of mutant clones, and the Simpson's Index of mutant clones. Utilizing these quantities, we developed a set of estimators for the model parameters: the growth rate of sensitive cells (λ_0), the growth rate of resistant cells (λ_1), the birth rate of resistant cells (r_1), and the mutation rate ($n^{-\alpha}$). These parameters are crucial for the development of personalized treatment plans for patients and in quantifying the strength of the selective pressure induced by the introduction of the anti-cancer drug or antibiotic. Through simulation, we demonstrated that these estimators exhibit robust performance across a wide range of parameter settings, underscoring their potential utility in clinical and pharmaceutical research.

In contrast to our prior work (Leder and Wang 2021), this paper examines a more general model wherein resistant cancer cells are modeled using a birth-death process, replacing the Yule process employed previously. A key contribution of this paper is the establishment of several convergence in probability results with explicitly specified convergence rates. These results enable the use of data from a single patient to construct estimators that are provably consistent, a significant improvement over our earlier approach that relied on convergence in expectation results and necessitated multiple

data points from different patients for estimation. In practice, it's important to note that even for the same type of tumors, individual variability factors such as immune system response, genetic differences, and treatment history can affect cell fitness and mutation rates. While our previous estimators offered a general overview of these parameters, the estimators developed in this study provide individualized information, thereby facilitating more targeted medical advice. Furthermore, this paper developed an estimator for the birth rate of resistant cells (r_1), a variable that was not investigated in Leder and Wang (2021).

This study has several limitations that could serve as potential directions for future research. Based on our simulation results, the estimators $\hat{\lambda}_0^{(n)}$, $\hat{\lambda}_1^{(n)}$, and $\hat{\alpha}^{(n)}$ demonstrate very good performance, whereas $\hat{r}_1^{(n)}$ does not exhibit equivalent performance. This observation aligns with that made by Gunnarsson et al. (2023), who reported lesser accuracy in estimating the birth rate compared to the growth rate. They suggested that this discrepancy may be attributed to the fact that the birth rate can only be estimated using higher order moments. While we have established the convergence of $\hat{r}_1^{(n)}$ to r_1 , the rate of this convergence remains undetermined. Our results suggest that, depending on the specific parameter settings, the initial burden n might need to increase to as large as 10^{11} for error reduction to commence. Yet, our simulations indicate a trend of decline in estimation errors when the initial burden n is above 10^5 . In addition, the bootstrapping procedure that we created significantly improves the accuracy of $\hat{r}_1^{(n)}$. In future work, we plan to establish a stronger theoretical result regarding the convergence of $\hat{r}_1^{(n)}$ and theoretically explain the improved performance of $\hat{r}_1^{(n)}$ due to the bootstrapping procedure.

Another limitation of our study is the assumption that all resistant cells have the same fitness level. However, this assumption is potentially oversimplified, as highlighted by Elowitz et al. (2002), Feinerman et al. (2008), which document random gene expressions in genetically identical cancer cells. In order to consider this aspect, both (Foo et al. 2014) and (Durrett et al. 2011a) employ a similar multi-type branching process model, introducing heterogeneity among resistant cells by assuming random fitness effects for newly mutated cells. In "Appendix G", we conduct a simulation experiment to test the performance of our estimators when the fitness of resistant cells follows a Gaussian distribution. The simulation results indicate that the performance of our estimators, particularly those for λ_1 and r_1 , is negatively affected by the random fitness effect, especially when the variance of the distribution is high. Expanding on this idea, an extension of our work could involve the development of estimators for models that incorporate these random fitness effects. A particularly interesting extension would be to design a non-parametric estimation framework to estimate the distribution of these random fitness changes.

A third limitation of our current model is the assumption that drug resistance in cancer cells is driven by a single mutational event. While this assumption holds true in certain scenarios, such as the T790M mutation in non-small cell lung cancer conferring resistance to erlotinib (Suda et al. 2009), there are cases where multiple mutations are required for cancer cells to develop drug resistance. For example, gene amplification, which allows cancer cells to evade targeted therapies (Ilic et al. 2011), necessitate a more complex model. To accommodate these scenarios, our model could be expanded

to account for multiple types of cancer cells, classified based on the number of relevant mutations they possess.

Lastly, our model does not account for the possibility of pre-existing drug resistance in cancer cells. An interesting extension to the current work would involve leveraging sequencing data obtained at the point of diagnosis and at recurrence. This approach would enable the development of refined models capable of estimating parameters that govern the dynamics of pre-existing resistant cells.

Supplement

The source code for the simulation experiments is available on GitHub at the following URL: <https://github.com/zhan8093/Single-Sample-Estimation.git>.

A Preliminary asymptotic results

For the sake of simplicity and conciseness in the subsequent proofs, this section lists the magnitudes of several important quantities we will frequently use. For the main model, the deterministic limit of the cancer recurrence time, ζ_n , and the variance of the size of resistant cells at the deterministic limit have the following orders:

$$\zeta_n \sim \frac{\alpha}{\lambda_1} \log n, \quad (\text{A.1})$$

$$\text{Var} [Z_1^n(\zeta_n)] \sim n^{1+\alpha}. \quad (\text{A.2})$$

Denote by $Z(t)$ the population size of a birth-death process starting from a single cell with birth rate r , death rate d , and growth rate $\lambda = r - d$. We have

$$\mathbb{E} [Z(t)^2] = \frac{2r}{\lambda} e^{2\lambda t} - \frac{r+d}{\lambda} e^{\lambda t}. \quad (\text{A.3})$$

Denote by $Z^n(t)$ the population size of a birth-death process starting from n cells with the same birth/death rate specified previously. We have

$$\mathbb{E} [Z^n(t)^2] = n \frac{r+d}{\lambda} e^{\lambda t} (e^{\lambda t} - 1) + n^2 e^{2\lambda t}. \quad (\text{A.4})$$

Moreover, from Foo et al. (2014), we have

$$\mathbb{E} [Z^n(t)^k] = \Theta \left(n^k e^{k\lambda t} \right). \quad (\text{A.5})$$

B Proof of Theorem 1

Proof We first show the following lemma, which is an extension of Foo and Leder (2013), Lemma 2.

Lemma 1 For any $b > 0$ and $u < (1 - \alpha)/2$, we have

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\sup_{0 \leq k \leq b} n^{\frac{\lambda_1 k}{\lambda_0} - u} (Z_1^n(kt_n) - \Phi_1^n(kt_n)) > \epsilon \right) = 0,$$

where $t_n = -\frac{1}{\lambda_0} \log n$, and $\Phi_1^n(\cdot)$ is defined in (2.3).

Proof The proof of Lemma 1 is similar to that of Foo and Leder (2013), Lemma 2 with a slight change in the scaling, and thus we omit the details here.

To prove Theorem 1, we can apply the same arguments in the proof of Hanagal et al. (2022), Theorem 1 with Lemma 1 and a slight change in the scaling. Therefore, we omit the details here for brevity. $\square$

C Proof of Theorem 2

We first establish a convergence result for the number of mutant clones at the deterministic limit of cancer recurrence time (i.e., ζ_n) in Lemma 2.

Lemma 2 For any $\epsilon > 0$, $u < \min\{(1 - \alpha)/2, -\lambda_0\alpha/2\lambda_1\}$, we have

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(n^u \left| \frac{1}{n^{1-\alpha}} I_n(\zeta_n) + \frac{\lambda_1}{\lambda_0 r_1} \right| > \epsilon \right) = 0. \quad (\text{C.1})$$

Proof Conditioned on $\{Z_0^n(t), t \leq \zeta_n\}$, $I_n(t)$ is a non-homogeneous Poisson process with arrival rate $\frac{\lambda_1}{r_1} n^{-\alpha} Z_0^n(t)$ for $t \leq \zeta_n$. Therefore, we have

$$\begin{aligned} \mathbb{E} [I_n(\zeta_n)^2] &= \mathbb{E} \left[\mathbb{E} [I_n(\zeta_n)^2 | Z_0^n(t), t \leq \zeta_n] \right] \\ &= \mathbb{E} \left[\left(\int_0^{\zeta_n} \frac{\lambda_1}{r_1} n^{-\alpha} Z_0^n(t) dt \right)^2 + \int_0^{\zeta_n} \frac{\lambda_1}{r_1} n^{-\alpha} Z_0^n(t) dt \right] \\ &= \frac{\lambda_1^2}{r_1^2} n^{-2\alpha} \mathbb{E} \left[\left(\int_0^{\zeta_n} Z_0^n(t) dt \right)^2 \right] + \frac{\lambda_1}{r_1} n^{1-\alpha} \int_0^{\zeta_n} e^{\lambda_0 s} ds. \end{aligned} \quad (\text{C.2})$$

For the expected value in the first term of (C.2), we have

$$\mathbb{E} \left[\left(\int_0^{\zeta_n} Z_0^n(t) dt \right)^2 \right] \quad (\text{C.3})$$

$$\stackrel{(a)}{=} \int_0^{\zeta_n} \int_0^{\zeta_n} \mathbb{E} [Z_0^n(t) Z_0^n(s)] dt ds \quad (\text{C.4})$$

$$= \int_0^{\zeta_n} \int_s^{\zeta_n} \mathbb{E} [Z_0^n(t) Z_0^n(s)] dt ds + \int_0^{\zeta_n} \int_0^s \mathbb{E} [Z_0^n(t) Z_0^n(s)] dt ds, \quad (\text{C.5})$$

where we use Fubini's Theorem in equality (a). For the first term in (C.5), we can apply (A.4) and obtain that

$$\begin{aligned}
 & \int_0^{\zeta_n} \int_s^{\zeta_n} \mathbb{E} [Z_0^n(t) Z_0^n(s)] dt ds \\
 &= \int_0^{\zeta_n} \int_s^{\zeta_n} \mathbb{E} [\mathbb{E} [Z_0^n(t) Z_0^n(s) | Z_0^n(s)]] dt ds \\
 &= \int_0^{\zeta_n} \int_s^{\zeta_n} \mathbb{E} [Z_0^n(s)^2 e^{\lambda_0(t-s)}] dt ds \\
 &= \int_0^{\zeta_n} \int_s^{\zeta_n} e^{\lambda_0(t-s)} \left(-n \frac{r_0 + d_0}{\lambda_0} e^{\lambda_0 s} (1 - e^{\lambda_0 s}) + n^2 e^{2\lambda_0 s} \right) dt ds \\
 &= \int_0^{\zeta_n} -\frac{1}{\lambda_0} (1 - e^{\lambda_0(s-\zeta_n)}) \left(-n \frac{r_0 + d_0}{\lambda_0} e^{\lambda_0 s} (1 - e^{\lambda_0 s}) + n^2 e^{2\lambda_0 s} \right) ds.
 \end{aligned}$$

Similarly, for the second term in (C.5), we can obtain that

$$\begin{aligned}
 & \int_0^{\zeta_n} \int_0^s \mathbb{E} [Z_0^n(t) Z_0^n(s)] dt ds \\
 &= \int_0^{\zeta_n} \int_0^s \mathbb{E} [Z_0^n(t)^2 e^{\lambda_0(s-t)}] dt ds \\
 &= \int_0^{\zeta_n} \int_0^s e^{\lambda_0(s-t)} \left(-n \frac{r_0 + d_0}{\lambda_0} e^{\lambda_0 t} (1 - e^{\lambda_0 t}) + n^2 e^{2\lambda_0 t} \right) dt ds \\
 &= \int_0^{\zeta_n} \left(-n \frac{r_0 + d_0}{\lambda_0} e^{\lambda_0 s} \left(s - \frac{1}{\lambda_0} e^{\lambda_0 s} + \frac{1}{\lambda_0} \right) + n^2 e^{\lambda_0 s} \left(\frac{1}{\lambda_0} e^{\lambda_0 s} - \frac{1}{\lambda_0} \right) \right) ds.
 \end{aligned}$$

Combining these two terms, we have

$$\begin{aligned}
 & \mathbb{E} \left[\left(\int_0^{\zeta_n} Z_0^n(t) dt \right)^2 \right] - \frac{n^2}{\lambda_0^2} \\
 &= \int_0^{\zeta_n} n^2 e^{\lambda_0 s} \left(-\frac{1}{\lambda_0} + \frac{1}{\lambda_0} e^{\lambda_0 \zeta_n} \right) ds \\
 &\quad - \int_0^{\zeta_n} n \frac{r_0 + d_0}{\lambda_0} e^{\lambda_0 s} \left(s - \frac{1}{\lambda_0} e^{\lambda_0 \zeta_n} + \frac{1}{\lambda_0} e^{-\lambda_0(s-\zeta_n)} \right) ds - \frac{n^2}{\lambda_0^2} \\
 &\sim C_1 n^2 e^{\lambda_0 \zeta_n} + C_2 n,
 \end{aligned}$$

where C_1 and C_2 are some constants, and we use (A.1) in the last step. Hence, we can get

$$n^{2\alpha-2} \mathbb{E} [I_n(\zeta_n)^2] - \frac{\lambda_1^2}{\lambda_0^2 r_1^2} \sim -\frac{\lambda_1}{\lambda_0 r_1} n^{\alpha-1} + C_1 n^{\alpha\lambda_0/\lambda_1} + C_2 n^{-1}.$$

Then we use (2.7), Markov's Inequality, and the condition $u < \min\{(1 - \alpha)/2, -\alpha\lambda_0/2\lambda_1\}$ to obtain that

$$\mathbb{P}\left(n^u \left| n^{\alpha-1} I_n(\zeta_n) + \frac{\lambda_1}{\lambda_0 r_1} \right| > \epsilon\right) \leq n^{2u} \mathbb{E} \left[\left(n^{\alpha-1} I_n(\zeta_n) + \frac{\lambda_1}{\lambda_0 r_1} \right)^2 \right] / \epsilon^2 \rightarrow 0.$$

□

Next, we show that the number of mutant clones at the cancer recurrence time (i.e., $I_n(\gamma_n)$) is close to that at the deterministic limit time (i.e., $I_n(\zeta_n)$).

Lemma 3 For any $\epsilon > 0$, $u < (1 - \alpha)/2 - \alpha\lambda_0/\lambda_1$, we have

$$\lim_{n \rightarrow \infty} \mathbb{P}\left(n^u \left| \frac{1}{n^{1-\alpha}} (I_n(\zeta_n) - I_n(\gamma_n)) \right| > \epsilon\right) = 0.$$

Proof For any $\delta > 0$, let $\delta_n = \delta n^{-k}$ where $k < (1 - \alpha)/2$. We have

$$\begin{aligned} & \mathbb{P}\left(n^u \left| \frac{1}{n^{1-\alpha}} (I_n(\zeta_n) - I_n(\gamma_n)) \right| > \epsilon\right) \\ & \leq \mathbb{P}\left(n^u \left| \frac{1}{n^{1-\alpha}} (I_n(\zeta_n) - I_n(\gamma_n)) \right| > \epsilon, |\zeta_n - \gamma_n| < \delta_n\right) + \mathbb{P}(|\zeta_n - \gamma_n| > \delta_n). \end{aligned}$$

It suffices to consider the first term as the second term goes to zero by Theorem 1 and the assumption that $k < (1 - \alpha)/2$. Because $I_n(t)$ is a counting process which is non-decreasing in t , when $|\zeta_n - \gamma_n| < \delta_n$, we have

$$\left| \frac{1}{n^{1-\alpha}} (I_n(\zeta_n) - I_n(\gamma_n)) \right| \leq \frac{1}{n^{1-\alpha}} (I_n(\zeta_n + \delta_n) - I_n(\zeta_n - \delta_n)).$$

Therefore,

$$\begin{aligned} & \mathbb{P}\left(n^u \left| \frac{1}{n^{1-\alpha}} (I_n(\zeta_n) - I_n(\gamma_n)) \right| > \epsilon, |\zeta_n - \gamma_n| < \delta_n\right) \\ & \leq \mathbb{P}\left(\frac{1}{n^{1-\alpha-u}} (I_n(\zeta_n + \delta_n) - I_n(\zeta_n - \delta_n)) > \epsilon\right) \\ & \leq n^{\alpha+u-1} \mathbb{E} [I_n(\zeta_n + \delta_n) - I_n(\zeta_n - \delta_n)] / \epsilon \\ & = \frac{\lambda_1}{\epsilon r_1} n^{u-1} \mathbb{E} \left[\int_{\zeta_n - \delta_n}^{\zeta_n + \delta_n} Z_0^n(t) dt \right] \\ & = \frac{\lambda_1}{\epsilon r_1} n^{u-1} \int_{\zeta_n - \delta_n}^{\zeta_n + \delta_n} \mathbb{E} [Z_0^n(t)] dt \\ & = -\frac{\lambda_1}{\epsilon \lambda_0 r_1} n^u \left(e^{\lambda_0(\zeta_n - \delta_n)} - e^{\lambda_0(\zeta_n + \delta_n)} \right) = \Theta(n^{\lambda_0\alpha/\lambda_1+u} \delta_n). \end{aligned}$$

Therefore, when $u < (1 - \alpha)/2 - \lambda_0\alpha/\lambda_1$, we can find $k < (1 - \alpha)/2$ such that

$$\limsup_{n \rightarrow \infty} \mathbb{P} \left(n^u \left| \frac{1}{n^{1-\alpha}} (I_n(\zeta_n) - I_n(\gamma_n)) \right| > \epsilon, |\zeta_n - \gamma_n| < \delta_n \right) = 0,$$

which implies that

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(n^u \left| \frac{1}{n^{1-\alpha}} (I_n(\zeta_n) - I_n(\gamma_n)) \right| > \epsilon \right) = 0.$$

□

proof of Theorem 2 Theorem 2 follows directly from Lemmas 2 and 3. □

D Proof of Theorem 3

For ease of analysis, we redefine the Simpson's Index $R_n(t)$ by considering the number of all mutant clones generated before time t , which we denote by $\tilde{I}_n(t)$. The formula of the Simpson's Index becomes

$$R_n(t) = \sum_{i=1}^{\tilde{I}_n(t)} \left(\frac{X_{i,n}(t)}{Z_1^n(t)} \right)^2.$$

This change of definition does not change the value of $R_n(t)$ as we only add some clones with size 0 at time t .

In Lemma 4, Corollaries 2 and 3, we present a large deviation result on the size of resistant cells at a scale of cancer recurrence time. For ease of notation, we define

$$f(x) = \frac{r_1}{\lambda_1} (1 - e^{-x}), \quad g(x, s) = \frac{r_1}{\lambda_1} e^{-x} e^{-\lambda_1(\zeta_n - s)}, \quad h(s) = \frac{d_1}{\lambda_1} e^{-\lambda_1(\zeta_n - s)}$$

Lemma 4 For any $\epsilon > 0$, $\alpha < 1$, and any (l, k) such that $k < (1 - \lambda_0/\lambda_1)\alpha$, $l > k$, and $l + k < \min\{1 - \alpha, \alpha\}$, there exists $c > 0$ so that

$$\limsup_{n \rightarrow \infty} \frac{1}{n^{1-\alpha-(l+k)}} \log \mathbb{P} (Z_1^n(\zeta_n) < (1 - \epsilon_n)n) \leq -c,$$

where $\epsilon_n = \epsilon n^{-k}$.

Remark: We can always find a pair of (l, k) such that the conditions specified in Lemma 4 are met. For example, we can let $l = \min\{(1 - \alpha)/2, \alpha/2\}$ and choose $k < \min\{(1 - \alpha)/2, \alpha/2\}$.

Proof Similar to the arguments in the proof of Leder and Wang (2021), Proposition 1, we have

$$\log \mathbb{P} (Z_1^n(\zeta_n) < (1 - \epsilon_n)n)$$

$$\leq \min_{x_n > 0} \left[x_n (1 - \epsilon_n) n + \frac{1}{n^\alpha} \int_0^{\zeta_n} n e^{\lambda_0 s} \left(\frac{e^{-x_n} - 1}{f(x_n) + g(x_n, s) - h(s)} \right) ds \right] \quad (\text{D.1})$$

$$+ k_1 (\log n)^2 n^{1-2\alpha}, \quad (\text{D.2})$$

where (D.2) comes from Hanagal et al. (2022), Proposition 1 and $k_1 > 0$ is some constant. Fix $x_n = \delta_n e^{-\lambda_1 \zeta_n}$ where $\delta_n = \delta n^{-l}$, and since we are now dropping the minimization operator we will have an upper bound for (D.1). From the definition of ζ_n and (2.3), we have

$$e^{-\lambda_1 \zeta_n} = \frac{1}{\lambda_1 - \lambda_0} n^{-\alpha} \left(1 - e^{(\lambda_0 - \lambda_1) \zeta_n} \right).$$

We can then obtain that the inner part of (D.1) will be less than

$$\begin{aligned} & \frac{\delta_n (1 - \epsilon_n)}{\lambda_1 - \lambda_0} n^{1-\alpha} - n^{1-\alpha} \int_0^{\zeta_n} e^{\lambda_0 s} \frac{x_n}{\frac{r_1}{\lambda_1} x_n + e^{-\lambda_1 (\zeta_n - s)}} ds \\ & + n^{1-\alpha} \int_0^{\zeta_n} e^{\lambda_0 s} \left(\frac{e^{-x_n} - 1}{f(x_n) + g(x_n, s) - h(s)} + \frac{x_n}{\frac{r_1}{\lambda_1} x_n + e^{-\lambda_1 (\zeta_n - s)}} \right) ds \\ & - \frac{\delta_n (1 - \epsilon_n)}{\lambda_1 - \lambda_0} n^{1-\alpha} e^{(\lambda_0 - \lambda_1) \zeta_n} \\ & \leq \frac{\delta_n (1 - \epsilon_n)}{\lambda_1 - \lambda_0} n^{1-\alpha} - n^{1-\alpha} \int_0^{\zeta_n} e^{\lambda_0 s} \frac{x_n}{\frac{r_1}{\lambda_1} x_n + e^{-\lambda_1 (\zeta_n - s)}} ds \quad (*) \\ & + n^{1-\alpha} \int_0^{\zeta_n} e^{\lambda_0 s} \left(\frac{e^{-x_n} - 1}{f(x_n) + g(x_n, s) - h(s)} + \frac{x_n}{\frac{r_1}{\lambda_1} x_n + e^{-\lambda_1 (\zeta_n - s)}} \right) ds. \quad (**) \end{aligned}$$

For the first term, we have

$$\begin{aligned} (*) &= \frac{\delta_n (1 - \epsilon_n)}{\lambda_1 - \lambda_0} n^{1-\alpha} - n^{1-\alpha} \int_0^{\zeta_n} e^{\lambda_0 s} \frac{\delta_n}{\frac{r_1}{\lambda_1} \delta_n + e^{\lambda_1 s}} ds \\ &= n^{1-\alpha-l} \left(\frac{\delta}{\lambda_1 - \lambda_0} (1 - \epsilon_n) - \int_0^{\zeta_n} e^{\lambda_0 s} \frac{\delta}{\frac{r_1}{\lambda_1} \delta_n + e^{\lambda_1 s}} ds \right) \\ &= \delta n^{1-\alpha-l} \left(-\frac{\epsilon_n}{\lambda_1 - \lambda_0} + \int_0^\infty \frac{e^{\lambda_0 s}}{e^{\lambda_1 s}} ds - \int_0^{\zeta_n} \frac{e^{\lambda_0 s}}{\frac{r_1}{\lambda_1} \delta_n + e^{\lambda_1 s}} ds \right) \\ &= \delta n^{1-\alpha-l} \left(-\frac{\epsilon_n}{\lambda_1 - \lambda_0} + \frac{e^{(\lambda_0 - \lambda_1) \zeta_n}}{\lambda_1 - \lambda_0} + \int_0^{\zeta_n} \left(\frac{e^{\lambda_0 s}}{e^{\lambda_1 s}} - \frac{e^{\lambda_0 s}}{\frac{r_1}{\lambda_1} \delta_n + e^{\lambda_1 s}} \right) ds \right) \\ &= \delta n^{1-\alpha-l} \left(-\frac{\epsilon_n}{\lambda_1 - \lambda_0} + \frac{e^{(\lambda_0 - \lambda_1) \zeta_n}}{\lambda_1 - \lambda_0} + \frac{r_1}{\lambda_1} \delta_n \int_0^{\zeta_n} \frac{e^{(\lambda_0 - \lambda_1) s}}{\frac{r_1}{\lambda_1} \delta_n + e^{\lambda_1 s}} ds \right). \end{aligned}$$

By (A.1), we have

$$e^{(\lambda_0 - \lambda_1)\zeta_n} = \Theta(n^{-(1 - \lambda_0/\lambda_1)\alpha}), \quad \delta_n = \Theta(n^{-l}), \quad \text{and} \quad \epsilon_n = \Theta(n^{-k}).$$

Therefore, when $l > k$ and $k < (1 - \lambda_0/\lambda_1)\alpha$, $(*)$ will be dominated by the term $-\frac{\delta\epsilon_n}{\lambda_1 - \lambda_0}n^{1-\alpha-l}$ and thus $(*) = \Theta(n^{1-\alpha-(l+k)})$ and is negative. Meanwhile, for the second term, we have

$$(**) = n^{1-\alpha} \int_0^{\zeta_n} e^{\lambda_0 s} \frac{\left(\frac{r_1}{\lambda_1} x_n e^{-x_n} - 1 - \frac{d_1}{\lambda_1} x_n + e^{-x_n}\right) e^{-\lambda_1(\zeta_n - s)}}{\left(\frac{r_1}{\lambda_1} x_n + e^{-\lambda_1(\zeta_n - s)}\right) (f(x_n) + g(x_n, s) - h(s))} ds. \quad (\text{D.3})$$

Because $r_1 > d_1$ and x_n converges to 0, by the Taylor expansion of e^{-x_n} , we can obtain that the numerator in (D.3) is negative and the denominator is positive when n is sufficiently large. Therefore we can get an upper bound of the absolute value of $(**)$ as follows.

$$\begin{aligned} |(**)| &\leq n^{1-\alpha} \int_0^{\zeta_n} e^{\lambda_0 s} \frac{\left(\left(\frac{r_1}{\lambda_1} - \frac{1}{2}\right) x_n^2 + \frac{1}{6} x_n^3\right) e^{-\lambda_1(\zeta_n - s)}}{\left(\frac{r_1}{\lambda_1} x_n + e^{-\lambda_1(\zeta_n - s)}\right) \left(\frac{r_1}{\lambda_1} \left(x_n - \frac{x_n^2}{2}\right) + \frac{r_1}{\lambda_1} (1 - x_n) e^{-\lambda_1(\zeta_n - s)} - h(s)\right)} \\ &\quad ds \sim \left(\frac{r_1}{\lambda_1} - \frac{1}{2}\right) n^{1-\alpha} \delta_n^2 e^{-\lambda_1 \zeta_n} \int_0^{\zeta_n} \frac{e^{(\lambda_0 + \lambda_1)s}}{e^{2\lambda_1 s}} ds = \Theta(n^{1-2\alpha-2l}). \end{aligned}$$

Because $l > k$ and $\alpha > 0$, we have $1 - 2\alpha - 2l < 1 - \alpha - (l + k)$ which implies that $(**)$ is dominated by $(*)$. Finally, because $l + k < \alpha$, (D.2) is dominated by $(*)$ as well, which leads to the desired result such that for some $c > 0$, we have

$$\limsup_{n \rightarrow \infty} \frac{1}{n^{1-\alpha-(l+k)}} \log \mathbb{P}(Z_1^n(\zeta_n) < (1 - \epsilon_n)n) \leq -c.$$

□

Corollary 2 For any $\epsilon > 0$, $\alpha < 1$, and any (l, k) such that $k < (1 + r/\lambda_1)\alpha$, $l > k$ and $l + k < \min\{1 - \alpha, \alpha\}$, there exists $c > 0$ so that

$$\limsup_{n \rightarrow \infty} \frac{1}{n^{1-\alpha-(l+k)}} \log \mathbb{P}(Z_1^n(\zeta_n) > (1 + \epsilon_n)n) \leq -c,$$

where $\epsilon_n = \epsilon n^{-k}$.

Proof Similar to the proof of Lemma 4, we can obtain that

$$\begin{aligned} &\log \mathbb{P}(Z_1^n(\zeta_n) > (1 + \epsilon_n)n) \\ &\leq \min_{0 < x_n} \left[-x_n (1 + \epsilon_n)n + n^{1-\alpha} \int_0^{\zeta_n} e^{\lambda_0 s} \left(\frac{e^{x_n} - 1}{f(-x_n) + g(-x_n, s) - h(s)} \right) ds \right] \end{aligned} \quad (\text{D.4})$$

$$+ k_1 (\log n)^2 n^{1-2\alpha}. \quad (\text{D.5})$$

Let $x_n = \delta_n e^{-\lambda_1 \zeta_n}$ where $\delta_n = \delta n^{-l}$. We can obtain that the inner part of (D.4) is equal to

$$\begin{aligned} & -x_n (1 + \epsilon_n) n + n^{1-\alpha} \int_0^{\zeta_n} e^{\lambda_0 s} \left(\frac{e^{x_n} - 1}{f(-x_n) + g(-x_n, s) - h(s)} \right) ds \\ &= -\frac{\delta_n (1 + \epsilon_n)}{\lambda_1 - \lambda_0} n^{1-\alpha} + n^{1-\alpha} \int_0^{\zeta_n} e^{\lambda_0 s} \frac{\delta_n}{-\frac{r_1}{\lambda_1} \delta_n + e^{\lambda_1 s}} ds \quad (*) \\ &+ n^{1-\alpha} \int_0^{\zeta_n} e^{\lambda_0 s} \left(\frac{e^{x_n} - 1}{f(-x_n) + g(-x_n, s) - h(s)} \right) ds \quad (**) \\ &+ \frac{\delta_n (1 + \epsilon_n)}{\lambda_1 - \lambda_0} n^{1-\alpha} e^{(\lambda_0 - \lambda_1) \zeta_n}. \quad (***) \end{aligned}$$

For the first term, we have

$$\begin{aligned} (*) &= -\frac{\delta_n (1 + \epsilon_n)}{\lambda_1 - \lambda_0} n^{1-\alpha} + n^{1-\alpha} \int_0^{\zeta_n} e^{\lambda_0 s} \frac{\delta_n}{-\frac{r_1}{\lambda_1} \delta_n + e^{\lambda_1 s}} ds \\ &= n^{1-\alpha-l} \left(-\frac{\delta}{\lambda_1 - \lambda_0} (1 + \epsilon_n) + \int_0^{\zeta_n} e^{\lambda_0 s} \frac{\delta}{-\frac{r_1}{\lambda_1} \delta_n + e^{\lambda_1 s}} ds \right) \\ &= \delta n^{1-\alpha-l} \left(-\frac{\epsilon_n}{\lambda_1 - \lambda_0} - \int_0^{\infty} \frac{e^{\lambda_0 s}}{e^{\lambda_1 s}} ds + \int_0^{\zeta_n} \frac{e^{\lambda_0 s}}{-\frac{r_1}{\lambda_1} \delta_n + e^{\lambda_1 s}} ds \right) \\ &= \delta n^{1-\alpha-l} \left(-\frac{\epsilon_n}{\lambda_1 - \lambda_0} - \frac{e^{(\lambda_0 - \lambda_1) \zeta_n}}{\lambda_1 - \lambda_0} - \int_0^{\zeta_n} \left(\frac{e^{\lambda_0 s}}{e^{\lambda_1 s}} - \frac{e^{\lambda_0 s}}{-\frac{r_1}{\lambda_1} \delta_n + e^{\lambda_1 s}} \right) ds \right) \\ &= \delta n^{1-\alpha-l} \left(-\frac{\epsilon_n}{\lambda_1 - \lambda_0} - \frac{e^{(\lambda_0 - \lambda_1) \zeta_n}}{\lambda_1 - \lambda_0} - \frac{r_1}{\lambda_1} \delta_n \int_0^{\zeta_n} \frac{e^{(\lambda_0 - \lambda_1) s}}{-\frac{r_1}{\lambda_1} \delta_n + e^{\lambda_1 s}} ds \right). \end{aligned}$$

When $l > k$ and $k < (1 + r/\lambda_1)\alpha$, we have $(*) = \Theta(n^{1-\alpha-l-k})$. The analysis of the second term and the third term are similar to that in Lemma 4 and we can obtain that

$$\begin{aligned} (**) &= \Theta(n^{1-2\alpha-2l}) \\ (***) &= \Theta(n^{1-\alpha-l-(1+r/\lambda_1)\alpha}), \end{aligned}$$

which are both dominated by the first term. When $l+k < \alpha$, the (D.5) is also dominated by (*). Therefore, for some $c > 0$, we have

$$\limsup_{n \rightarrow \infty} \frac{1}{n^{1-\alpha-(l+k)}} \log \mathbb{P} (Z_1^n (\zeta_n) > (1 + \epsilon_n) n) \leq -c.$$

□

Corollary 3 For any $\epsilon > 0$, $\alpha < 1$ and (l, k) satisfying $1 < k < l < k + \frac{1-\alpha}{\alpha}$, there exists $c > 0$ so that

$$\limsup_{n \rightarrow \infty} \frac{1}{n^{1-(l-k+1)\alpha}} \log \mathbb{P}(Z_1^n(k\zeta_n) < n) \leq -c,$$

Proof For further ease of notation, we define

$$g_k(x, s) = \frac{r_1}{\lambda_1} e^{-x} e^{-\lambda_1(k\zeta_n-s)}, \quad h_k(s) = \frac{d_1}{\lambda_1} e^{-\lambda_1(k\zeta_n-s)}$$

Similar with the proof of the Lemma 4, we have

$$\begin{aligned} & \log \mathbb{P}(Z_1^n(\zeta_n) < (1 - \epsilon_n)n) \\ & \leq \min_{x_n > 0} \left[x_n n + \frac{1}{n^\alpha} \int_0^{k\zeta_n} n e^{\lambda_0 s} \left(\frac{e^{-x_n} - 1}{f(x_n) + g_k(x_n, s) - h_k(s)} \right) ds \right] \end{aligned} \quad (\text{D.6})$$

$$+ k_1 (\log n)^2 n^{1-2\alpha}, \quad (\text{D.7})$$

Let $x_n = \delta e^{-l\lambda_1 \zeta_n}$ where l can be any number satisfying $k < l < k + \frac{1-\alpha}{\alpha}$. We can obtain

$$\begin{aligned} & n^{(l-k+1)\alpha-1} \left(x_n n + \frac{1}{n^\alpha} \int_0^{k\zeta_n} n e^{\lambda_0 s} \left(\frac{e^{-x_n} - 1}{f(x_n) + g_k(x_n, s) - h_k(s)} \right) ds \right) \\ & = n^{(l-k)\alpha} \left(\delta n^\alpha e^{-l\lambda_1 \zeta_n} + \int_0^{k\zeta_n} e^{\lambda_0 s} \left(\frac{e^{-x_n} - 1}{f(x_n) + g_k(x_n, s) - h_k(s)} \right) ds \right) \\ & = n^{(l-k)\alpha} \left(\delta n^\alpha e^{-l\lambda_1 \zeta_n} - e^{k\lambda_0 \zeta_n} \int_0^{k\zeta_n} e^{(\lambda_1 - \lambda_0)s} \left(\frac{1 - e^{-x_n}}{f(x_n) e^{\lambda_1 s} + \frac{r_1}{\lambda_1} e^{-x_n} - \frac{d_1}{\lambda_1}} \right) ds \right) \\ & = n^{(l-k)\alpha} \left(\delta n^\alpha e^{-l\lambda_1 \zeta_n} - e^{k\lambda_0 \zeta_n} (1 - e^{-x_n}) \int_0^{k\zeta_n} \left(\frac{e^{(\lambda_1 - \lambda_0)(k\zeta_n-s)}}{f(x_n) e^{\lambda_1(k\zeta_n-s)} + \frac{r_1}{\lambda_1} e^{-x_n} - \frac{d_1}{\lambda_1}} \right) ds \right) \\ & = n^{(l-k)\alpha} \left(\delta n^\alpha e^{-l\lambda_1 \zeta_n} - e^{k\lambda_1 \zeta_n} (1 - e^{-x_n}) \int_0^{k\zeta_n} \left(\frac{e^{-(\lambda_1 - \lambda_0)s}}{f(x_n) e^{\lambda_1(k\zeta_n-s)} + \frac{r_1}{\lambda_1} e^{-x_n} - \frac{d_1}{\lambda_1}} \right) ds \right) \end{aligned}$$

By Dominated Convergence Theorem, we have

$$\lim_{n \rightarrow \infty} \int_0^{k\zeta_n} \left(\frac{e^{-(\lambda_1 - \lambda_0)s}}{\frac{r_1}{\lambda_1} (1 - e^{-x_n}) e^{\lambda_1(k\zeta_n-s)} + \frac{r_1}{\lambda_1} e^{-x_n} - \frac{d_1}{\lambda_1}} \right) ds = \int_0^\infty e^{-(\lambda_1 - \lambda_0)s} ds = \frac{1}{\lambda_1 - \lambda_0}$$

Because of $1 - e^{-x_n} \sim x_n = \delta n^{-l\alpha}$ and $e^{k\lambda_1 \zeta_n} \sim (\lambda_1 - \lambda_0)^k n^{k\alpha}$, we can find $c > 0$ and

$$\limsup_{n \rightarrow \infty} n^{(l-k+1)\alpha-1} \left(x_n n + \frac{1}{n^\alpha} \int_0^{k\zeta_n} n e^{\lambda_0 s} \left(\frac{e^{-x_n} - 1}{f(x_n) + g_k(x_n, s) - h_k(s)} \right) ds \right) < -c$$

□

In Lemma 5, we show a convergence in probability result for the Simpson's Index of mutant clones at the deterministic limit of cancer recurrence time. For ease of exposition, we define $\mathbb{P}_{\rho_n}(A) = \mathbb{P}(A \cap \rho_n)$ and $\mathbb{E}_{\rho_n}[X] = \mathbb{E}[X \cdot 1_{\rho_n}]$.

Lemma 5 *For any $\epsilon > 0$, when $u < \min\{\alpha/4, (1 - \alpha)/4\}$, we have*

$$\lim_{n \rightarrow \infty} \mathbb{P}_{\rho_n} \left(n^u \left| n^{1-\alpha} R_n(\zeta_n) - \frac{2r_1(\lambda_1 - \lambda_0)^2}{\lambda_1(2\lambda_1 - \lambda_0)} \right| > \epsilon \right) = 0.$$

Proof By Chebyshev's inequality, we have

$$\begin{aligned} & \mathbb{P}_{\rho_n} \left(n^u \left| n^{1-\alpha} R_n(\zeta_n) - \frac{2r_1(\lambda_1 - \lambda_0)^2}{\lambda_1(2\lambda_1 - \lambda_0)} \right| > \epsilon \right) \\ & \leq \frac{n^{2-2\alpha+2u}}{\epsilon^2} \mathbb{E}_{\rho_n} \left[\left(R_n(\zeta_n) - n^{\alpha-1} \frac{2r_1(\lambda_1 - \lambda_0)^2}{\lambda_1(2\lambda_1 - \lambda_0)} \right)^2 \right] \\ & \leq \frac{n^{2-2\alpha+2u}}{\epsilon^2} \mathbb{E} \left[\left(R_n(\zeta_n) - n^{\alpha-1} \frac{2r_1(\lambda_1 - \lambda_0)^2}{\lambda_1(2\lambda_1 - \lambda_0)} \right)^2 \right]. \end{aligned}$$

Combined with (2.6), it suffices to show that

$$\lim_{n \rightarrow \infty} \frac{n^{2-2\alpha+2u}}{\epsilon^2} \left(\mathbb{E} \left[R_n(\zeta_n)^2 \right] - n^{2\alpha-2} \frac{4r_1^2(\lambda_1 - \lambda_0)^4}{\lambda_1^2(2\lambda_1 - \lambda_0)^2} \right) = 0.$$

Define

$$\tilde{R}_n(\zeta_n) = \sum_{i=1}^{\tilde{I}_n(\zeta_n)} \left(\frac{X_{i,n}(\zeta_n)}{n} \right)^2. \quad (\text{D.8})$$

where $\tilde{I}_n(t)$ represents the number of all mutant clones generated before time t . We can obtain that

$$\begin{aligned} n^{2u} \left| \mathbb{E} \left[R_n(\zeta_n)^2 \right] - \mathbb{E} \left[\tilde{R}_n(\zeta_n)^2 \right] \right| & \leq n^{2u} \mathbb{E} \left[\left| R_n(\zeta_n)^2 - \tilde{R}_n(\zeta_n)^2 \right| \right] \\ & = n^{2u} \mathbb{E}_{\rho_n} \left[\left| R_n(\zeta_n)^2 - \tilde{R}_n(\zeta_n)^2 \right| \right] \\ & \leq n^{2u} \mathbb{E}_{\rho_n} \left[\left| \tilde{R}_n(\zeta_n)^2 \left(\frac{n^4}{(Z_1^n(\zeta_n))^4} - 1 \right) \right| \right]. \end{aligned}$$

We know $\mathbb{P}_{\rho_n}(Z_1^n(\zeta_n) < (1 - \delta_n)n)$ and $\mathbb{P}_{\rho_n}(Z_1^n(\zeta_n) > (1 + \delta_n)n)$ decay exponentially for any $\delta_n = \delta n^{-k}$ where $k < \min\{(1 - \alpha)/2, \alpha/2\}$ according to Lemma 4 and

Corollary 2. Therefore,

$$\begin{aligned}
 & n^{2u} \mathbb{E}_{\rho_n} \left[\left| \tilde{R}_n(\zeta_n)^2 \left(\frac{n^4}{(Z_1^n(\zeta_n))^4} - 1 \right) \right| \right] \\
 &= n^{2u} \mathbb{E}_{\rho_n} \left[\left| \tilde{R}_n(\zeta_n)^2 \left(\frac{n^4}{(Z_1^n(\zeta_n))^4} - 1 \right) \right|, Z_1^n(\zeta_n) < (1 - \delta_n)n \right] \\
 &\quad + n^{2u} \mathbb{E}_{\rho_n} \left[\left| \tilde{R}_n(\zeta_n)^2 \left(\frac{n^4}{(Z_1^n(\zeta_n))^4} - 1 \right) \right|, Z_1^n(\zeta_n) \geq (1 + \delta_n)n \right] \\
 &\quad + n^{2u} \mathbb{E}_{\rho_n} \left[\left| \tilde{R}_n(\zeta_n)^2 \left(\frac{n^4}{(Z_1^n(\zeta_n))^4} - 1 \right) \right|, (1 + \delta_n)n > Z_1^n(\zeta_n) \geq (1 - \delta_n)n \right] \\
 &\leq n^{4+2u} \mathbb{E}_{\rho_n} \left[\tilde{R}_n(\zeta_n)^2 | Z_1^n(\zeta_n) < (1 - \delta_n)n \right] \mathbb{P}_{\rho_n}(Z_1^n(\zeta_n) < (1 - \delta_n)n) \\
 &\quad + n^{2u} \mathbb{E}_{\rho_n} \left[\frac{Z_1^n(\zeta_n)^4}{n^4}, Z_1^n(\zeta_n) \geq (1 + \delta_n)n \right] \\
 &\quad + n^{2u} \left(1/(1 - \delta_n)^4 - 1 \right) \mathbb{E}_{\rho_n} \left[\tilde{R}_n(\zeta_n)^2 \right] \\
 &\stackrel{(a)}{\leq} n^{4+2u} \mathbb{P}_{\rho_n}(Z_1^n(\zeta_n) < (1 - \delta_n)n) + n^{2u} \left(1/(1 - \delta_n)^4 - 1 \right) \mathbb{E}_{\rho_n} \left[\tilde{R}_n(\zeta_n)^2 \right] \\
 &\quad + n^{2u-4} \mathbb{E}_{\rho_n} \left[Z_1^n(\zeta_n)^8 \right]^{1/2} \mathbb{P}_{\rho_n}(Z_1^n(\zeta_n) \geq (1 + \delta_n)n)^{1/2} \\
 &\stackrel{(b)}{\sim} n^{2u} \left(1/(1 - \delta_n)^4 - 1 \right) \mathbb{E}_{\rho_n} \left[\tilde{R}_n(\zeta_n)^2 \right] \\
 &\sim 4n^{2u} \delta_n \mathbb{E}_{\rho_n} \left[\tilde{R}_n(\zeta_n)^2 \right],
 \end{aligned}$$

where we use Cauchy-Schwarz inequality for (a) and the following fact in (b):

$$\mathbb{E}_{\rho_n} \left[Z_1^n(\zeta_n)^8 \right] \leq Cn^8, \text{ for some constant } C > 0. \quad (\text{D.9})$$

To prove (D.9), we first observe that

$$\begin{aligned}
 \mathbb{E}_{\rho_n} \left[Z_1^n(\zeta_n)^8 \right] &= \mathbb{E} \left[Z_1^n(\zeta_n)^8 \right] \\
 &\stackrel{(a)}{\leq} \mathbb{E} \left[\tilde{I}_n(\zeta_n)^8 Z_1(\zeta_n)^8 \right] \\
 &= \mathbb{E} \left[\tilde{I}_n(\zeta_n)^8 \right] \mathbb{E} \left[Z_1(\zeta_n)^8 \right], \quad (\text{D.10})
 \end{aligned}$$

where $Z_1(t)$ is the population size of a branching process starting from a single resistant cell with birth rate r_1 and death rate d_1 , and the inequality (a) holds because $Z_1^n(\zeta_n)$ is bounded by the process assuming all the clones are generated at $t = 0$. For the first

term in (D.10), we have

$$\begin{aligned}\mathbb{E}\left[\tilde{I}_n(\zeta_n)^8\right] &= \mathbb{E}\left[\mathbb{E}\left[\tilde{I}_n(\zeta_n)^8 \mid Z_0^n(t), t \leq \zeta_n\right]\right] \\ &\stackrel{(a)}{=} \Theta\left(\mathbb{E}\left[\left(\int_0^{\zeta_n} n^{-\alpha} Z_0^n(t) dt\right)^8\right]\right),\end{aligned}$$

where we use the fact that the eighth moment of a Poisson distribution has the same order as the eighth power of its mean in equality (a). We can further show that

$$\begin{aligned}&\mathbb{E}\left[\left(\int_0^{\zeta_n} n^{-\alpha} Z_0^n(t) dt\right)^8\right] \\ &= n^{-8\alpha} \int_0^{\zeta_n} \cdots \int_0^{\zeta_n} \mathbb{E}\left[Z_0^n(x_1) \cdots Z_0^n(x_8)\right] dx_1 \cdots dx_8 \\ &\stackrel{(a)}{\leq} n^{-8\alpha} \left(\int_0^{\zeta_n} \mathbb{E}\left[Z_0^n(s)^8\right]^{1/8} ds\right)^8 \\ &= \Theta(n^{8-8\alpha}),\end{aligned}$$

where we use Cauchy–Schwarz Inequality repeatedly in inequality (a). As can be seen in Foo et al. (2014), Lemma 5, we have $\mathbb{E}\left[Z_1(\zeta_n)^8\right] = \Theta(n^{8\alpha})$, which completes the proof of (D.9). To conclude, we have obtained that

$$n^{2u} \left| \mathbb{E}\left[R_n(\zeta_n)^2\right] - \mathbb{E}\left[\tilde{R}_n(\zeta_n)^2\right] \right| = \Theta(4n^{2u} \delta_n \mathbb{E}_{\rho_n}\left[\tilde{R}_n(\zeta_n)^2\right]). \quad (\text{D.11})$$

From the proof of Leder and Wang (2021), Proposition 1, we can obtain that

$$\mathbb{E}_{\rho_n}\left[\tilde{R}_n(\zeta_n)^2\right] = \frac{4r_1^2(\lambda_1 - \lambda_0)^4}{\lambda_1^2(2\lambda_1 - \lambda_0)^2} n^{-2+2\alpha} + C_1 n^{-2+\alpha} + C_2 n^{-3+3\alpha} + o(n^{-2+\alpha}), \quad (\text{D.12})$$

where C_1 and C_2 are some constants. With (D.11) and (D.12), we have

$$\begin{aligned}&n^{2-2\alpha+2u} \left(\left| \mathbb{E}\left[R_n(\zeta_n)^2\right] - n^{2\alpha-2} \frac{4r_1^2(\lambda_1 - \lambda_0)^4}{\lambda_1^2(2\lambda_1 - \lambda_0)^2} \right| \right) \\ &\leq n^{2-2\alpha+2u} \left(\left| \mathbb{E}\left[R_n(\zeta_n)^2\right] - \mathbb{E}\left[\tilde{R}_n(\zeta_n)^2\right] \right| + \left| \mathbb{E}\left[\tilde{R}_n(\zeta_n)^2\right] - n^{2\alpha-2} \frac{4r_1^2(\lambda_1 - \lambda_0)^4}{\lambda_1^2(2\lambda_1 - \lambda_0)^2} \right| \right) \\ &= \Theta\left(4n^{2u} \delta_n n^{2-2\alpha} \mathbb{E}_{\rho_n}\left[\tilde{R}_n(\zeta_n)^2\right] + n^{2u-\alpha} + n^{2u-1+\alpha}\right).\end{aligned}$$

Therefore, when $u < \min\{\alpha/4, (1-\alpha)/4\}$, we have

$$\lim_{n \rightarrow \infty} \mathbb{P}_{\rho_n} \left(n^u \left| n^{1-\alpha} R_n(\zeta_n) - \frac{2r_1(\lambda_1 - \lambda_0)^2}{\lambda_1(2\lambda_1 - \lambda_0)} \right| > \epsilon \right) = 0.$$

□

In Lemma 6, we show that the Simpson's Index at the cancer recurrence time is very close to that at the deterministic limit time of cancer recurrence.

Lemma 6 For any $\epsilon > 0$, when $u < (1 - \alpha)/4$, we have

$$\lim_{n \rightarrow \infty} \mathbb{P}_{\rho_n} \left(n^u \left| n^{1-\alpha} R_n(\gamma_n) - n^{1-\alpha} R_n(\zeta_n) \right| > \epsilon \right) = 0.$$

Proof When $\zeta_n < \gamma_n$, we have

$$\begin{aligned} & \mathbb{P}_{\rho_n} \left(n^u \left| n^{1-\alpha} (R_n(\gamma_n) - R_n(\zeta_n)) \right| > \epsilon \right) \\ &= \mathbb{P}_{\rho_n} \left(n^u \left| n^{1-\alpha} \left(\sum_{i=1}^{\tilde{I}_n(\gamma_n)} \frac{X_{i,n}(\gamma_n)^2}{n^2} - \sum_{i=1}^{\tilde{I}_n(\zeta_n)} \frac{X_{i,n}(\zeta_n)^2}{Z_1^n(\zeta_n)^2} \right) \right| > \epsilon \right) \\ &\leq \mathbb{P}_{\rho_n} \left(n^u \left| n^{1-\alpha} \left(\sum_{i=1}^{\tilde{I}_n(\zeta_n)} \frac{X_{i,n}(\gamma_n)^2}{n^2} - \sum_{i=1}^{\tilde{I}_n(\zeta_n)} \frac{X_{i,n}(\zeta_n)^2}{Z_1^n(\zeta_n)^2} \right) \right| > \epsilon/2 \right) \\ &\quad + \mathbb{P}_{\rho_n} \left(\sum_{i=\tilde{I}_n(\zeta_n)+1}^{\tilde{I}_n(\gamma_n)} \frac{X_{i,n}(\gamma_n)^2}{n^{1+\alpha-u}} > \epsilon/2 \right) \\ &\leq \mathbb{P}_{\rho_n} \left(n^u \left| n^{1-\alpha} \left(\sum_{i=1}^{\tilde{I}_n(\zeta_n)} \frac{X_{i,n}(\gamma_n)^2}{n^2} - \sum_{i=1}^{\tilde{I}_n(\zeta_n)} \frac{X_{i,n}(\zeta_n)^2}{Z_1^n(\zeta_n)^2} \right) \right| > \epsilon/2 \right) \end{aligned} \quad (\text{D.13})$$

$$+ \mathbb{P} \left(\sum_{i=\tilde{I}_n(\zeta_n)+1}^{\tilde{I}_n(\gamma_n)} \frac{X_{i,n}(\gamma_n)^2}{n^{1+\alpha-u}} > \epsilon/2 \right). \quad (\text{D.14})$$

Here we define $X_{i,n}(t) = 0$ if the creation time of i_{th} clone is after t . Therefore, when $\zeta_n \geq \gamma_n$, we have

$$\begin{aligned} & \mathbb{P}_{\rho_n} \left(n^u \left| n^{1-\alpha} (R_n(\gamma_n) - R_n(\zeta_n)) \right| > \epsilon \right) \\ &= \mathbb{P}_{\rho_n} \left(n^u \left| n^{1-\alpha} \left(\sum_{i=1}^{\tilde{I}_n(\gamma_n)} \frac{X_{i,n}(\gamma_n)^2}{n^2} - \sum_{i=1}^{\tilde{I}_n(\zeta_n)} \frac{X_{i,n}(\zeta_n)^2}{Z_1^n(\zeta_n)^2} \right) \right| > \epsilon \right) \\ &= \mathbb{P}_{\rho_n} \left(n^u \left| n^{1-\alpha} \left(\sum_{i=1}^{\tilde{I}_n(\zeta_n)} \frac{X_{i,n}(\gamma_n)^2}{n^2} - \sum_{i=\tilde{I}_n(\gamma_n)+1}^{\tilde{I}_n(\zeta_n)} \frac{X_{i,n}(\gamma_n)^2}{n^2} \right. \right. \right. \\ &\quad \left. \left. \left. - \sum_{i=1}^{\tilde{I}_n(\zeta_n)} \frac{X_{i,n}(\zeta_n)^2}{Z_1^n(\zeta_n)^2} \right) \right| > \epsilon/2 \right) \end{aligned} \quad (\text{D.15})$$

Because of our definition above, we have $X_{i,n}(\gamma_n) = 0$ when $\tilde{I}_n(\gamma_n) + 1 \leq i \leq \tilde{I}_n(\zeta_n)$. Therefore, (D.15) can be reduced to (D.13).

For (D.14), we can obtain that

$$\begin{aligned} & \mathbb{P} \left(\sum_{i=\tilde{I}_n(\zeta_n)+1}^{\tilde{I}_n(\gamma_n)} \frac{X_{i,n}(\gamma_n)^2}{n^{1+\alpha-u}} > \epsilon/2 \right) \\ & \leq \mathbb{P} \left(\sum_{i=\tilde{I}_n(\zeta_n)+1}^{\tilde{I}_n(\gamma_n)} \frac{X_{i,n}(\gamma_n)^2}{n^{1+\alpha-u}} > \epsilon/2, \gamma_n \leq \zeta_n + \delta_n \right) + \mathbb{P}(\gamma_n > \zeta_n + \delta_n), \end{aligned}$$

where δ_n is some constant, and the summation is zero if $\zeta_n > \gamma_n$. From Theorem 1, we know that when $\delta_n = n^{-v}$, $v < (1 - \alpha)/2$, $\mathbb{P}(\gamma_n > \zeta_n + \delta_n)$ converges to 0. Moreover, we have

$$\begin{aligned} & \mathbb{P} \left(\sum_{i=\tilde{I}_n(\zeta_n)+1}^{\tilde{I}_n(\gamma_n)} \frac{X_{i,n}(\gamma_n)^2}{n^{1+\alpha-u}} > \epsilon/2, \gamma_n \leq \zeta_n + \delta_n \right) \\ & \leq \mathbb{P} \left(\sum_{i=\tilde{I}_n(\zeta_n)+1}^{\tilde{I}_n(\zeta_n+\delta_n)} \frac{X_{i,n}(\gamma_n)^2}{n^{1+\alpha-u}} > \epsilon/2, \gamma_n \leq \zeta_n + \delta_n \right) \\ & \leq \mathbb{P} \left(\sum_{i=\tilde{I}_n(\zeta_n)+1}^{\tilde{I}_n(\zeta_n+\delta_n)} \frac{Z^{(i)}(\delta_n)^2}{n^{1+\alpha-u}} > \epsilon/2 \right), \end{aligned}$$

where $Z^{(i)}(t)$'s are i.i.d copies of $Z(t)$, which is the population size of a death-birth process starting from a single resistant cell. By Markov's inequality and

$$\begin{aligned} \mathbb{E} \left[\tilde{I}_n(\zeta_n + \delta_n) - \tilde{I}_n(\zeta_n) \right] &= n^{1-\alpha} \int_{\zeta_n}^{\zeta_n+\delta_n} e^{\lambda_0 s} ds \\ &\sim n^{1-\alpha} e^{\lambda_0 \zeta_n} \delta_n \sim n^{1-\alpha+\lambda_0 \alpha / \lambda_1 - v}, \end{aligned}$$

we have

$$\begin{aligned} \mathbb{P} \left(\sum_{i=\tilde{I}_n(\zeta_n)+1}^{\tilde{I}_n(\zeta_n+\delta_n)} \frac{Z^{(i)}(\delta_n)^2}{n^{1+\alpha-u}} > \epsilon/2 \right) &\leq \frac{2}{\epsilon n^{1+\alpha-u}} \mathbb{E} \left[\tilde{I}_n(\zeta_n + \delta_n) - \tilde{I}_n(\zeta_n) \right] \mathbb{E} \left[Z(\delta_n)^2 \right] \\ &= \Theta(n^{u-2\alpha-v+\lambda_0 \alpha / \lambda_1}) \rightarrow 0. \end{aligned}$$

Next we analyze (D.13). We have

$$\begin{aligned} & \mathbb{P}_{\rho_n} \left(n^u \left| n^{1-\alpha} \left(\sum_{i=1}^{\tilde{I}_n(\zeta_n)} \frac{X_{i,n}(\gamma_n)^2}{n^2} - \sum_{i=1}^{\tilde{I}_n(\zeta_n)} \frac{X_{i,n}(\zeta_n)^2}{Z_1^n(\zeta_n)^2} \right) \right| > \epsilon/2 \right) \\ & \leq \mathbb{P} \left(n^u \left| n^{1-\alpha} \left(\sum_{i=1}^{\tilde{I}_n(\zeta_n)} \frac{X_{i,n}(\gamma_n)^2}{n^2} - \sum_{i=1}^{\tilde{I}_n(\zeta_n)} \frac{X_{i,n}(\zeta_n)^2}{n^2} \right) \right| > \epsilon/2 \right) \end{aligned} \quad (\text{D.16})$$

$$+ \mathbb{P}_{\rho_n} \left(n^u \left| n^{1-\alpha} \left(\sum_{i=1}^{\tilde{I}_n(\zeta_n)} \frac{X_{i,n}(\zeta_n)^2}{n^2} - \sum_{i=1}^{\tilde{I}_n(\zeta_n)} \frac{X_{i,n}(\zeta_n)^2}{Z_1^n(\zeta_n)^2} \right) \right| > \epsilon/2 \right). \quad (\text{D.17})$$

We work on (D.17) first and aim to show that

$$\lim_{n \rightarrow \infty} \mathbb{P}_{\rho_n} \left(n^u \left| n^{1-\alpha} \left(\sum_{i=1}^{\tilde{I}_n(\zeta_n)} \frac{X_{i,n}(\zeta_n)^2}{n^2} - \sum_{i=1}^{\tilde{I}_n(\zeta_n)} \frac{X_{i,n}(\zeta_n)^2}{Z_1^n(\zeta_n)^2} \right) \right| > \epsilon \right) = 0. \quad (\text{D.18})$$

For the term within the probability, we have

$$\begin{aligned} & n^{1-\alpha+u} \left| \sum_{i=1}^{\tilde{I}_n(\zeta_n)} \frac{X_{i,n}(\zeta_n)^2}{n^2} - \sum_{i=1}^{\tilde{I}_n(\zeta_n)} \frac{X_{i,n}(\zeta_n)^2}{Z_1^n(\zeta_n)^2} \right| \\ &= n^{1-\alpha+u} \left| \frac{1}{n^2} - \frac{1}{Z_1^n(\zeta_n)^2} \right| \sum_{i=1}^{\tilde{I}_n(\zeta_n)} X_{i,n}(\zeta_n)^2 \\ &= n^u \left(1 - \frac{n^2}{Z_1^n(\zeta_n)^2} \right) n^{1-\alpha} \sum_{i=1}^{\tilde{I}_n(\zeta_n)} n^{-2} X_{i,n}(\zeta_n)^2 \\ &= n^u \left(1 - \frac{n^2}{Z_1^n(\zeta_n)^2} \right) n^{1-\alpha} \tilde{R}_n(\zeta_n) \end{aligned}$$

In the proof of Lemma 5, we have shown that $n^{1-\alpha} \tilde{R}_n(\zeta_n)$ converges to a constant in probability. Moreover, by a direct calculation of $\mathbb{E}[Z_1^n(\zeta_n)^4]$ (see the proof of Leder and Wang (2021), Proposition 1), we have $\text{Var} \left(n^u \left(1 - \frac{Z_1^n(\zeta_n)^2}{n^2} \right) \right) = \Theta(n^{4\alpha-4+2u})$. Because $u < (1-\alpha)/4$, we have

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(n^u \left| 1 - \frac{Z_1^n(\zeta_n)^2}{n^2} \right| > \epsilon \right) = 0.$$

By some algebra, we have when $n^u \left| 1 - \frac{Z_1^n(\zeta_n)^2}{n^2} \right| < \epsilon$,

$$\frac{-\epsilon}{1-\epsilon n^{-u}} < n^u \left(1 - \frac{n^2}{Z_1^n(\zeta_n)^2} \right) < \frac{\epsilon}{1+\epsilon n^{-u}},$$

which implies

$$\lim_{n \rightarrow \infty} \mathbb{P}_{\rho_n} \left(n^u \left| 1 - \frac{n^2}{Z_1^n(\zeta_n)^2} \right| > \epsilon \right) = 0.$$

Therefore, by the continuous mapping theorem, $n^u \left(1 - \frac{n^2}{Z_n^u(\zeta_n)^2}\right) n^{1-\alpha} \tilde{R}_n(\zeta_n)$ converges to 0 in probability and we have shown the desired result in (D.18).

Next, we work on (D.16) and aim to show that

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(n^u \left| n^{1-\alpha} \left(\sum_{i=1}^{\tilde{I}_n(\zeta_n)} \frac{X_{i,n}(\zeta_n)^2 - X_{i,n}(\gamma_n)^2}{n^2} \right) \right| > \epsilon \right) = 0.$$

For simplicity, we define the event $A_n = \{\omega \mid |\gamma_n - \zeta_n| < \delta_n\}$ where $\delta_n = \delta n^{-v}$ and δ is a fixed constant. By Theorem 1, we know that when $v < (1-\alpha)/2$, $\lim_{n \rightarrow \infty} \mathbb{P}(A_n^C) = 0$. Hence, we have

$$\begin{aligned} & \mathbb{P} \left(n^u \left| n^{1-\alpha} \left(\sum_{i=1}^{\tilde{I}_n(\zeta_n)} \frac{X_{i,n}(\zeta_n)^2 - X_{i,n}(\gamma_n)^2}{n^2} \right) \right| > \epsilon \right) \\ & \leq \mathbb{P} \left(n^u \left| n^{1-\alpha} \left(\sum_{i=1}^{\tilde{I}_n(\zeta_n)} \frac{X_{i,n}(\zeta_n)^2 - X_{i,n}(\gamma_n)^2}{n^2} \right) \right| > \epsilon, A_n \right) + \mathbb{P}(A_n^C). \quad (\text{D.19}) \end{aligned}$$

For the first term in (D.19), we have

$$\begin{aligned} & \left| n^{1-\alpha+u} \left(\sum_{i=1}^{\tilde{I}_n(\zeta_n)} \frac{X_{i,n}(\zeta_n)^2 - X_{i,n}(\gamma_n)^2}{n^2} \right) \right| \cdot 1_{A_n} \\ & \leq n^{-1+\alpha+u} \sup_{0 \leq \Delta t \leq 2\delta_n} n^{-2\alpha} \left| \sum_{i=1}^{\tilde{I}_n(\zeta_n)} (X_{i,n}(\zeta_n)^2 - X_{i,n}(\zeta_n - \delta_n + \Delta t)^2) \right| \\ & \leq n^{-1+\alpha+u} \sum_{i=1}^{\tilde{I}_n(\zeta_n)} \sup_{0 \leq \Delta t \leq 2\delta_n} n^{-2\alpha} |X_{i,n}(\zeta_n)^2 - X_{i,n}(\zeta_n - \delta_n + \Delta t)^2| \\ & \leq n^{-1+\alpha+u} \sum_{i=1}^{\tilde{I}_n(\zeta_n)} n^{-2\alpha} |X_{i,n}(\zeta_n)^2 - X_{i,n}(\zeta_n - \delta_n)^2| \\ & \quad + n^{-1+\alpha+u} \sum_{i=1}^{\tilde{I}_n(\zeta_n)} \sup_{0 \leq \Delta t \leq 2\delta_n} n^{-2\alpha} |X_{i,n}(\zeta_n - \delta_n + \Delta t)^2 - X_{i,n}(\zeta_n - \delta_n)^2|. \end{aligned}$$

Therefore, by Markov's inequality,

$$\begin{aligned} & \mathbb{P} \left(n^u \left| n^{-1-\alpha} \left(\sum_{i=1}^{\tilde{I}_n(\zeta_n)} \frac{X_{i,n}(\zeta_n)^2 - X_{i,n}(\gamma_n)^2}{n^2} \right) \right| > \epsilon, A_n \right) \\ & \leq \frac{1}{\epsilon} \mathbb{E} \left[n^{-1+\alpha+u} \sum_{i=1}^{\tilde{I}_n(\zeta_n)} \sup_{0 \leq \Delta t \leq 2\delta_n} n^{-2\alpha} \left| X_{i,n}(\zeta_n - \delta_n + \Delta t)^2 - X_{i,n}(\zeta_n - \delta_n)^2 \right| \right] \end{aligned} \quad (\text{D.20})$$

$$+ \frac{1}{\epsilon} \mathbb{E} \left[n^{-1+\alpha+u} \sum_{i=1}^{\tilde{I}_n(\zeta_n)} n^{-2\alpha} \left| X_{i,n}(\zeta_n)^2 - X_{i,n}(\zeta_n - \delta_n)^2 \right| \right]. \quad (\text{D.21})$$

We bound (D.20) first. Note that

$$n^{-1+\alpha} \mathbb{E} [\tilde{I}_n(\zeta_n)] \rightarrow C,$$

where C is some constant. If we condition on the value of $\tilde{I}_n(\zeta_n)$, then by the uniformity property of Poisson process, the generation time of each clone is independently and identically distributed, and we denote by τ a random variable which follows such distribution. In addition, because $\tilde{I}_n(\zeta_n)$ and $X_{i,n}$ are independent, by Wald's equation, we have

$$\begin{aligned} & \frac{1}{\epsilon} \mathbb{E} \left[n^{-1+\alpha+u} \sum_{i=1}^{\tilde{I}_n(\zeta_n)} \sup_{0 \leq \Delta t \leq 2\delta_n} n^{-2\alpha} \left| X_{i,n}(\zeta_n - \delta_n + \Delta t)^2 - X_{i,n}(\zeta_n - \delta_n)^2 \right| \right] \\ & \sim \frac{C}{\epsilon} n^{-2\alpha+u} \mathbb{E} \left[\sup_{0 \leq \Delta t \leq 2\delta_n} \left| Z(\zeta_n - \delta_n - \tau + \Delta t)^2 - Z(\zeta_n - \delta_n - \tau)^2 \right| \right] \\ & \leq \frac{C}{\epsilon} n^{-2\alpha+u} \mathbb{E} \left[\left(\sup_{0 \leq \Delta t \leq 2\delta_n} \left| Z(\zeta_n - \delta_n - \tau + \Delta t)^2 - Z(\zeta_n - \delta_n - \tau)^2 \right| \right)^2 \right]^{1/2} \\ & = \frac{C}{\epsilon} n^{-2\alpha+u} \mathbb{E} \left[\sup_{0 \leq \Delta t \leq 2\delta_n} \left(Z(\zeta_n - \delta_n - \tau + \Delta t)^2 - Z(\zeta_n - \delta_n - \tau)^2 \right)^2 \right]^{1/2} \\ & = \frac{C}{\epsilon} n^{-2\alpha+u} \mathbb{E} \left[\mathbb{E} \left[\sup_{0 \leq \Delta t \leq 2\delta_n} \left(Z(\zeta_n - \delta_n - \tau + \Delta t)^2 - Z(\zeta_n - \delta_n - \tau)^2 \right)^2 \middle| \tau \right] \right]^{1/2}, \end{aligned}$$

where $Z(t)$ is the population size of a branching process starting from a single resistant cell. It's noteworthy that t can be less than 0 for $Z(t)$. As a natural extension from the definition of $X_{i,n}(t)$, we set $Z(t) = 0$ when $t < 0$. By Doob's Maximal Inequality

and the independence between τ and $Z(t)$, we have

$$\begin{aligned} & \frac{C}{\epsilon} n^{-2\alpha+u} \mathbb{E} \left[\mathbb{E} \left[\sup_{0 \leq \Delta t \leq 2\delta_n} \left(Z(\zeta_n - \delta_n - \tau + \Delta t)^2 - Z(\zeta_n - \delta_n - \tau)^2 \right)^2 \middle| \tau \right] \right]^{1/2} \\ & \leq \frac{4C}{\epsilon} n^{-2\alpha+u} \mathbb{E} \left[\mathbb{E} \left[\left(Z(\zeta_n - \tau + \delta_n)^2 - Z(\zeta_n - \tau - \delta_n)^2 \right)^2 \middle| \tau \right] \right]^{1/2} \\ & \stackrel{(a)}{\leq} \frac{4C}{\epsilon} n^{-2\alpha+u} \mathbb{E} \left[\mathbb{E} \left[Z(\zeta_n - \tau + \delta_n)^4 - Z(\zeta_n - \tau - \delta_n)^4 \middle| \tau \right] \right]^{1/2} \\ & \stackrel{(b)}{\leq} \frac{4C}{\epsilon} n^{-2\alpha+u} \mathbb{E} \left[Z(\zeta_n + \delta_n)^4 - Z(\zeta_n - \delta_n)^4 \right]^{1/2} \\ & \stackrel{(c)}{=} \Theta(n^{u-v/2}), \end{aligned}$$

where we use the fact that $Z(t)^2$ is a submartingale in inequality (a); $\mathbb{E}[Z(t)^4]$ is increasing in t in inequality (b); $\mathbb{E}[Z(t)^4] = \Theta(e^{4\lambda_1 t})$ and $e^{4\lambda_1 \zeta_n} = \Theta(n^{4\alpha})$ in equality (c).

Finally, we bound (D.21). By a similar argument, we have

$$\begin{aligned} & \frac{1}{\epsilon} \mathbb{E} \left[n^{-1+\alpha+u} \sum_{i=1}^{\tilde{I}_n(\zeta_n)} n^{-2\alpha} |X_{i,n}(\zeta_n)^2 - X_{i,n}(\zeta_n - \delta_n)^2| \right] \\ & \sim \frac{C}{\epsilon} n^{-2\alpha+u} \mathbb{E} \left[|Z(\zeta_n - \tau)^2 - Z(\zeta_n - \tau - \delta_n)^2| \right] \\ & \leq \frac{C}{\epsilon} n^{-2\alpha+u} \mathbb{E} \left[\left(Z(\zeta_n - \tau)^2 - Z(\zeta_n - \tau - \delta_n)^2 \right)^2 \right]^{1/2} \\ & \leq \frac{C}{\epsilon} n^{-2\alpha+u} \mathbb{E} \left[Z(\zeta_n - \tau)^4 + Z(\zeta_n - \tau - \delta_n)^4 - 2Z(\zeta_n - \tau - \delta_n)^2 Z(\zeta_n - \tau)^2 \right]^{1/2} \\ & \leq \frac{C}{\epsilon} n^{-2\alpha+u} \mathbb{E} \left[Z(\zeta_n - \tau)^4 - Z(\zeta_n - \tau - \delta_n)^4 \right]^{1/2} \\ & \leq \frac{C}{\epsilon} n^{-2\alpha+u} \mathbb{E} \left[Z(\zeta_n)^4 - Z(\zeta_n - \delta_n)^4 \right]^{1/2} \\ & = \Theta(n^{u-v/2}). \end{aligned}$$

Because $u < (1 - \alpha)/4$, we can find $v < (1 - \alpha)/2$ so that $u < v/2$ which, combined with the previous analysis, implies that

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(n^u \left| n^{1-\alpha} \left(\sum_{i=1}^{\tilde{I}_n(\zeta_n)} \frac{X_{i,n}(\zeta_n)^2 - X_{i,n}(\gamma_n)^2}{n^2} \right) \right| > \epsilon \right) = 0.$$

To conclude, when $u < (1 - \alpha)/4$ we have

$$\lim_{n \rightarrow \infty} \mathbb{P}_{\rho_n} \left(n^u \left| n^{1-\alpha} R_n(\gamma_n) - n^{1-\alpha} R_n(\zeta_n) \right| > \epsilon \right) = 0.$$

□

Proof of Theorem 3 Theorem 3 follows directly from Proposition 2, Lemmas 5 and 6. $\square$

E Proof of Theorem 4

Lemma 7 When $\alpha < -\frac{\lambda_1}{\lambda_0}$

$$\lim_{n \rightarrow \infty} \mathbb{P}(Z_0^n(\gamma_n) = 0) = 0$$

Proof

$$\begin{aligned} \mathbb{P}(Z_0^n(\gamma_n) = 0) &\leq \mathbb{P}(Z_0^n(\gamma_n) = 0, \gamma_n \leq \zeta_n + \epsilon) + \mathbb{P}(\gamma_n > \zeta_n + \epsilon) \\ &\leq \mathbb{P}(Z_0^n(\zeta_n + \epsilon) = 0) + \mathbb{P}(\gamma_n > \zeta_n + \epsilon) \end{aligned}$$

From Theorem 1 we know $\mathbb{P}(\gamma_n > \zeta_n + \epsilon) \rightarrow 0$ and we have

$$\begin{aligned} \mathbb{P}(Z_0^n(\zeta_n + \epsilon) = 0) &\leq \mathbb{P}(Z_0^n(\zeta_n + \epsilon) < ne^{\lambda_0(\zeta_n + \epsilon)}/2) \\ &\leq \mathbb{P}(|Z_0^n(\zeta_n + \epsilon) - ne^{\lambda_0(\zeta_n + \epsilon)}| > ne^{\lambda_0(\zeta_n + \epsilon)}/2) \\ &\leq 4\text{Var}(Z_0^n(\zeta_n + \epsilon))/n^2 e^{2\lambda_0(\zeta_n + \epsilon)} \end{aligned}$$

From (A.4), we know

$$\begin{aligned} 4\text{Var}(Z_0^n(\zeta_n + \epsilon))/n^2 e^{2\lambda_0(\zeta_n + \epsilon)} &= \frac{4n(r_0 + d_0)e^{\lambda_0(\zeta_n + \epsilon)}(e^{\lambda_0(\zeta_n + \epsilon)} - 1)}{\lambda_0 n^2 e^{2\lambda_0(\zeta_n + \epsilon)}} \\ &\sim \frac{1}{ne^{\lambda_0(\zeta_n + \epsilon)}} \rightarrow 0 \end{aligned}$$

$\square$

Proof of Theorem 4 We will prove the convergence of $\hat{\lambda}_0^{(n)}$, $\hat{\lambda}_1^{(n)}$, $\hat{\alpha}^{(n)}$ and $\hat{r}_1^{(n)}$ respectively.

- $\hat{\lambda}_0^{(n)}$: We show a stronger convergence result for $\hat{\lambda}_0^{(n)}$. That is for $\epsilon > 0$ and $u < \min\{(1 - \alpha)/2, (1 + \lambda_0\alpha/\lambda_1)/2\}$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(n^u |\hat{\lambda}_0^{(n)} - \lambda_0| > \epsilon) = 0.$$

Let $\epsilon_n = \epsilon n^{-v}$, where $u < v < (1 - \alpha)/2$ By Theorem 1 and Lemma 7 we have

$$\begin{aligned} \lim_{n \rightarrow \infty} \mathbb{P}(|\zeta_n - \gamma_n| > \epsilon_n) &= 0, \text{ and} \\ \lim_{n \rightarrow \infty} \mathbb{P}(Z_0^n(\gamma_n) = 0) &= 0. \end{aligned}$$

Define $A_n = \{\omega \mid |\zeta_n - \gamma_n| < \epsilon_n, Z_0^n(\gamma_n) > 0\}$ and we have $\lim_{n \rightarrow \infty} \mathbb{P}(A_n^C) = 0$. When $\omega \in A_n$, we have $\zeta_n - \epsilon_n < \gamma_n$ and $Z_0^n(\gamma_n) > 0$, and therefore we must have $Z_0^n(\zeta_n - \epsilon_n) > 0$ and

$$\begin{aligned} & n^u \left| \hat{\lambda}_0^{(n)} - \lambda_0 \right| \\ &= n^u \left| \frac{1}{\gamma_n} \log \frac{Z_0^n(\gamma_n)}{n} - \lambda_0 \right| \\ &= n^u \left| \frac{1}{\gamma_n} \log \frac{Z_0^n(\gamma_n)}{n} - \frac{1}{\gamma_n} \log \frac{Z_0^n(\zeta_n - \epsilon_n)}{n} \right. \\ &\quad \left. + \frac{1}{\gamma_n} \log \frac{Z_0^n(\zeta_n - \epsilon_n)}{n} - \frac{1}{\zeta_n} \log \frac{Z_0^n(\zeta_n - \epsilon_n)}{n} \right. \\ &\quad \left. + \frac{1}{\zeta_n} \log \frac{Z_0^n(\zeta_n - \epsilon_n)}{n} - \lambda_0 \right| \\ &\leq \frac{n^u}{\gamma_n} \left| \log \frac{Z_0^n(\gamma_n)}{Z_0^n(\zeta_n - \epsilon_n)} \right| + \frac{n^u \epsilon_n}{\gamma_n \zeta_n} \log \frac{n}{Z_0^n(\zeta_n - \epsilon_n)} + \frac{n^u}{\zeta_n} \left| \log \frac{Z_0^n(\zeta_n - \epsilon_n)}{n e^{\lambda_0 \zeta_n}} \right|. \end{aligned}$$

Because $\zeta_n = \Theta(\log n)$, $\zeta_n - \gamma_n$ converges to 0 in probability, and $\log \frac{n}{Z_0^n(\zeta_n - \epsilon_n)} < \log n$, the second term converges to 0 in probability. It remains to show that $\frac{n^u}{\log n} \left| \log \frac{Z_0^n(\gamma_n)}{Z_0^n(\zeta_n - \epsilon_n)} \right|$ and $\frac{n^u}{\log n} \left| \log \frac{Z_0^n(\zeta_n - \epsilon_n)}{n e^{\lambda_0 \zeta_n}} \right|$ converge to 0 in probability, which are equivalent to

$$\left(\left(\frac{Z_0^n(\gamma_n)}{Z_0^n(\zeta_n - \epsilon_n)} \vee \frac{Z_0^n(\zeta_n - \epsilon_n)}{Z_0^n(\gamma_n)} \right) - 1 + 1 \right)^{n^u / \log n} \xrightarrow{p} 1, \quad (\text{E.1})$$

$$\left(\left(\frac{Z_0^n(\zeta_n - \epsilon_n)}{n e^{\lambda_0 \zeta_n}} \vee \frac{n e^{\lambda_0 \zeta_n}}{Z_0^n(\zeta_n - \epsilon_n)} \right) - 1 + 1 \right)^{n^u / \log n} \xrightarrow{p} 1. \quad (\text{E.2})$$

Consider a non-negative random variable $X(n)$ such that there exists a positive constant C such that $\lim_{n \rightarrow \infty} \mathbb{P}(n^u X(n) \leq C) = 1$. Then on the event $\{n^u X(n) \leq C\}$, we have

$$\begin{aligned} & (X(n) + 1)^{n^u / \log n} \\ &= \left(\frac{n^u X(n)}{\log n} \frac{\log n}{n^u} + 1 \right)^{n^u / \log n} \\ &\leq \exp \left[\frac{C}{\log n} \right], \end{aligned}$$

which implies that $(X(n) + 1)^{n^u / \log n}$ converges to 1 in probability. With this result, to prove (E.1) and (E.2), it suffices to show that

$$n^u \left(\left(\frac{Z_0^n(\gamma_n)}{Z_0^n(\zeta_n - \epsilon_n)} \vee \frac{Z_0^n(\zeta_n - \epsilon_n)}{Z_0^n(\gamma_n)} \right) - 1 \right) \quad (\text{E.3})$$

and

$$n^u \left(\left(\frac{Z_0^n(\zeta_n - \epsilon_n)}{ne^{\lambda_0 \zeta_n}} \vee \frac{ne^{\lambda_0 \zeta_n}}{Z_0^n(\zeta_n - \epsilon_n)} \right) - 1 \right) \quad (\text{E.4})$$

are bounded by a constant in probability. We analyze (E.4) first. From (A.4) and the condition that $u < 1/2 + \lambda_0 \alpha / 2\lambda_1$, we have

$$\begin{aligned} \text{Var} \left(n^u \frac{Z_0^n(\zeta_n - \epsilon_n)}{ne^{\lambda_0 \zeta_n}} \right) &= \Theta \left(n^{2u-1} e^{-\lambda_0(\zeta_n + \epsilon_n)} \right) = \Theta \left(n^{-\frac{\lambda_0 \alpha}{\lambda_1} - 1 + 2u} \right) \rightarrow 0, \text{ and} \\ \mathbb{E} \left[n^u \left(\frac{Z_0^n(\zeta_n - \epsilon_n)}{ne^{\lambda_0 \zeta_n}} - 1 \right) \right] &= n^u (e^{-\lambda_0 \epsilon_n} - 1) \rightarrow 0. \end{aligned}$$

Therefore, $n^u (Z_0^n(\zeta_n - \epsilon_n)/ne^{\lambda_0 \zeta_n} - 1)$ converges to 0 in probability. By a similar argument to that in the proof of Lemma 6, we can obtain that (E.4) is bounded by a constant in probability.

We then analyze (E.3).

$$\begin{aligned} n^u \left(\frac{Z_0^n(\gamma_n)}{Z_0^n(\zeta_n - \epsilon_n)} - 1 \right) &= n^u \left(\frac{Z_0^n(\gamma_n)}{Z_0^n(\zeta_n - \epsilon_n)} - 1 \right) 1_{\{\gamma_n \in (\zeta_n - \epsilon_n, \zeta_n + \epsilon_n)\}} \\ &\quad + n^u \left(\frac{Z_0^n(\gamma_n)}{Z_0^n(\zeta_n - \epsilon_n)} - 1 \right) 1_{\{\gamma_n \notin (\zeta_n - \epsilon_n, \zeta_n + \epsilon_n)\}}. \end{aligned}$$

By Theorem 1, we can safely discard the second term. For the first term, we have

$$\begin{aligned} &n^u \left(\frac{Z_0^n(\gamma_n)}{Z_0^n(\zeta_n - \epsilon_n)} - 1 \right) 1_{\{\gamma_n \in (\zeta_n - \epsilon_n, \zeta_n + \epsilon_n)\}} \\ &\leq 1_{\{\gamma_n \in (\zeta_n - \epsilon_n, \zeta_n + \epsilon_n)\}} \frac{1}{Z_0^n(\zeta_n - \epsilon_n)} \sum_{i=1}^{Z_0^n(\zeta_n - \epsilon_n)} n^u \left(\sup_{t \in [0, 2\epsilon_n]} B_0^{(i)}(t) - 1 \right), \text{ and} \\ &n^u \left(\frac{Z_0^n(\gamma_n)}{Z_0^n(\zeta_n - \epsilon_n)} - 1 \right) 1_{\{\gamma_n \in (\zeta_n - \epsilon_n, \zeta_n + \epsilon_n)\}} \\ &\geq 1_{\{\gamma_n \in (\zeta_n - \epsilon_n, \zeta_n + \epsilon_n)\}} \frac{1}{Z_0^n(\zeta_n - \epsilon_n)} \sum_{i=1}^{Z_0^n(\zeta_n - \epsilon_n)} n^u \left(\inf_{t \in [0, 2\epsilon_n]} B_0^{(i)}(t) - 1 \right), \end{aligned}$$

where $\{B_0^{(i)}(t)\}$'s are i.i.d copies of the size of a branching process starting from a single sensitive cell; the upper bound is non-negative and the lower bound is non-positive. Notice that the right-continuous process $\{B_0^i(t)\}$'s are independent with $Z_0^n(\zeta_n - \epsilon_n)$. Hence, we have

$$\lim_{n \rightarrow \infty} \mathbb{E} \left[\frac{1}{Z_0^n(\zeta_n - \epsilon_n)} \sum_{i=1}^{Z_0^n(\zeta_n - \epsilon_n)} n^u \left(\sup_{t \in [0, 2\epsilon_n]} B_0^{(i)}(t) - 1 \right) \right]$$

$$\begin{aligned}
 &= \lim_{n \rightarrow \infty} \mathbb{E} \left[n^u \left(\sup_{t \in [0, 2\epsilon_n]} B_0^{(i)}(t) - 1 \right) \right] \\
 &\leq \lim_{n \rightarrow \infty} n^u \left(\mathbb{E} \left[\sup_{t \in [0, 2\epsilon_n]} \hat{B}_0^{(i)}(t) \right] - 1 \right) \\
 &= \lim_{n \rightarrow \infty} n^u \left(\mathbb{E} \left[\hat{B}_0^{(i)}(2\epsilon_n) \right] - 1 \right) \\
 &= \lim_{n \rightarrow \infty} n^u \left(e^{2r_0\epsilon_n} - 1 \right) \\
 &= 0,
 \end{aligned}$$

where $\hat{B}_0^{(i)}(t)$'s are i.i.d copies of the size of branching process starting from a cell with birth rate r_0 and death rate 0. Meanwhile,

$$\begin{aligned}
 &\lim_{n \rightarrow \infty} \mathbb{E} \left[\frac{1}{Z_0^n(\zeta_n - \epsilon_n)} \sum_{i=1}^{Z_0^n(\zeta_n - \epsilon_n)} n^u \left(1 - \inf_{t \in [0, 2\epsilon_n]} B_0^{(i)}(t) \right) \right] \\
 &= \lim_{n \rightarrow \infty} \mathbb{E} \left[n^u \left(1 - \inf_{t \in [0, 2\epsilon_n]} B_0^{(i)}(t) \right) \right] \\
 &\geq \lim_{n \rightarrow \infty} n^u \left(1 - \mathbb{E} \left[\inf_{t \in [0, 2\epsilon_n]} \tilde{B}_0^{(i)}(t) \right] \right) \\
 &= \lim_{n \rightarrow \infty} n^u \left(1 - \mathbb{E} \left[\tilde{B}_0^{(i)}(2\epsilon_n) \right] \right) \\
 &= \lim_{n \rightarrow \infty} n^u \left(1 - e^{-2d_0\epsilon_n} \right) \\
 &= 0,
 \end{aligned}$$

where $\tilde{B}_0^{(i)}(t)$'s are i.i.d copies of the size of branching process starting from a cell with birth rate 0 and death rate d_0 . Therefore, by Markov's inequality, the upper bound and lower bound of $n^u (Z_0^n(\gamma_n)/Z_0^n(\zeta_n - \epsilon_n) - 1)$ both converge to 0 in probability, which implies that $n^u (Z_0^n(\gamma_n)/Z_0^n(\zeta_n - \epsilon_n) - 1)$ converges to 0 in probability as well. Combined with the previous analysis, we have shown that

$$\lim_{n \rightarrow \infty} P \left(n^u \left| \hat{\lambda}_0^{(n)} - \lambda_0 \right| > \epsilon \right) = 0.$$

- $\hat{\lambda}_1^{(n)}$: Similar to $\hat{\lambda}_0^{(n)}$, we can prove a stronger convergence result for $\hat{\lambda}_1^{(n)}$. Specifically, we can show that for any $u < \min\{(1 - \alpha)/4, -\lambda_0\alpha/2\lambda_1, (1 + \lambda_0\alpha/\lambda_1)/2\}$ and $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(n^u \left| \hat{\lambda}_1^{(n)} - \lambda_1 \right| > \epsilon \right) = 0.$$

Recall that $U_n = \frac{\sqrt{I_n \cdot R_n}}{\sqrt{I_n \cdot R_n} - 2} - 1$. By Theorems 2 and 3, we can show that for any $v < \min\{(1 - \alpha)/4, -\lambda_0\alpha/2\lambda_1\}$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(n^v |U_n - U| > \epsilon) = 0, \quad (\text{E.5})$$

where $U = -\frac{\lambda_0}{\lambda_1}$. Note that

$$\begin{aligned} \hat{\lambda}_1^{(n)} - \lambda_1 &= -\frac{\hat{\lambda}_0^{(n)}}{U_n} + \frac{\hat{\lambda}_0^{(n)}}{U} - \frac{\hat{\lambda}_0^{(n)}}{U} + \frac{\lambda_0}{U} \\ &= -\hat{\lambda}_0^{(n)} \left(\frac{1}{U_n} - \frac{1}{U} \right) - \frac{1}{U} (\hat{\lambda}_0^{(n)} - \lambda_0). \end{aligned}$$

By (E.5) and the result on $\hat{\lambda}_0^{(n)}$ such that for any $w < \min\{(1 - \alpha)/2, (1 + \lambda_0\alpha/\lambda_1)/2\}$,

$$\lim_{n \rightarrow \infty} P(n^w |\hat{\lambda}_0^{(n)} - \lambda_0| > \epsilon) = 0,$$

we can obtain that for any $u < \min\{(1 - \alpha)/4, -\lambda_0\alpha/2\lambda_1, (1 + \lambda_0\alpha/\lambda_1)/2\}$, we have

$$\lim_{n \rightarrow \infty} P(n^u |\hat{\lambda}_1^{(n)} - \lambda_1| > \epsilon) = 0. \quad (\text{E.6})$$

- $\hat{r}_1^{(n)}$: Note that

$$\begin{aligned} \hat{r}_1^{(n)} &= \left(\frac{1}{U_n} + 1 \right) \frac{n\hat{\lambda}_1^{(n)}}{I_n e^{\hat{\lambda}_1^{(n)} \gamma_n}} \\ &= \left(\frac{1}{U_n} + 1 \right) \hat{\lambda}_1^{(n)} \frac{n^{1-\alpha}}{I_n} e^{\hat{\lambda}_1^{(n)} (\zeta_n - \gamma_n)} n^\alpha e^{-\hat{\lambda}_1^{(n)} \zeta_n} \\ &= \left(\frac{1}{U_n} + 1 \right) \hat{\lambda}_1^{(n)} \frac{n^{1-\alpha}}{I_n} e^{\hat{\lambda}_1^{(n)} (\zeta_n - \gamma_n)} (n^\alpha e^{-\lambda_1 \zeta_n}) e^{(\lambda_1 - \hat{\lambda}_1^{(n)}) \zeta_n}. \end{aligned}$$

By (E.5), (E.6), Theorems 2, 1, and (A.1), we can obtain that each term in the above expression $\left(\left(\frac{1}{U_n} + 1 \right), \hat{\lambda}_1^{(n)}, \frac{n^{1-\alpha}}{I_n}, e^{\hat{\lambda}_1^{(n)} (\zeta_n - \gamma_n)}, (n^\alpha e^{-\lambda_1 \zeta_n}), e^{(\lambda_1 - \hat{\lambda}_1^{(n)}) \zeta_n} \right)$ converges in probability to the corresponding limit $\left(\frac{\lambda_0 - \lambda_1}{\lambda_0}, \lambda_1, -\frac{\lambda_0 r_1}{\lambda_1}, 1, \frac{1}{\lambda_1 - \lambda_0}, 1 \right)$, respectively. Hence, $\hat{r}_1^{(n)}$ converges in probability to r_1 by the Continuous Mapping Theorem.

- $\hat{\alpha}_1^{(n)}$: By Theorem 2, we have

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\left| \frac{1}{n^{1-\alpha}} I_n(\gamma_n) + \frac{\lambda_1}{\lambda_0 r_1} \right| < \epsilon \right) = 1$$

Table 2 Simulation with a mutation rate of 10^{-3}

	γ_n	I_n	R_n
Deterministic limit	13.815	6666	4.000e-4
Average value	13.806	6711	3.996e-4
Average relative difference	0.0019	0.0126	0.0205
Problem parameters: $\lambda_0 = -0.5$, $\lambda_1 = 0.5$, $r_1 = 1.5$, $\alpha = 0.4286$, $n = 10^7$			

Table 3 Simulation with a mutation rate of 10^{-4}

	γ_n	I_n	R_n
Deterministic limit	18.420	666	4.0e-3
Average value	18.420	664	3.9e-3
Average relative difference	0.0047	0.0258	0.0460
Problem parameters: $\lambda_0 = -0.5$, $\lambda_1 = 0.5$, $r_1 = 1.5$, $\alpha = 0.5714$, $n = 10^7$			

which implies that

$$\lim_{n \rightarrow \infty} P \left(cn^{1-\alpha} \leq I_n(\gamma_n) \leq Cn^{1-\alpha} \right) = 1,$$

where c and C are some positive constants. It follows that

$$\hat{\alpha}^{(n)} - \alpha = 1 - \log_n I_n - \alpha + \log_n \left(\frac{\hat{\lambda}_1^{(n)}}{-\hat{\lambda}_0^{(n)} \hat{r}_1^{(n)}} \right) \xrightarrow{p} 0.$$

□

F Simulation with realistic resistance mutation rates

In this section, we conduct a simulation study using realistic resistance mutation rates to assess how closely the observable data - recurrence time (γ_n), number of mutant clones (I_n), and Simpson's Index (R_n) - align with their deterministic limits. Specifically, we perform the simulation with mutation rates of 10^{-3} , 10^{-4} , and 10^{-5} , which fall within the estimated range for drug resistance to cytotoxic drugs (see Duesberg et al. 2000 and references therein). The simulation results, presented in Table 2 through 4, involve performing 10 experiments for each parameter set. The results show that, for realistic resistance mutation rates, the observable data closely match their deterministic limits, as demonstrated in Theorems 1, 2, and 3 (Tables 3, 4).

G Simulation with random fitness effect

In our main model, we assume that all resistant cells have the same fitness level. In this section, we evaluate the performance of our estimators in scenarios where resistant

Table 4 Simulation with a mutation rate of 10^{-5}

	γ_n	I_n	R_n
Deterministic limit	23.025	66	4.0e-2
Average value	22.857	64	3.8e-2
Average relative difference	0.0168	0.1205	0.2004

Problem parameters: $\lambda_0 = -0.5$, $\lambda_1 = 0.5$, $r_1 = 1.5$, $\alpha = 0.7142$, $n = 10^7$

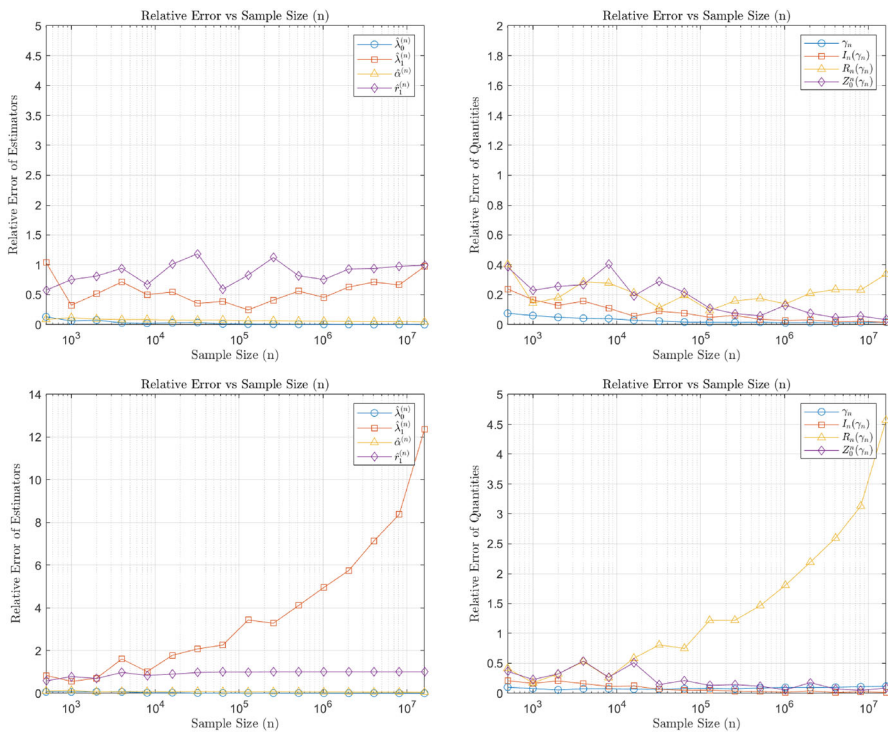


Fig. 6 In this experiment, $\alpha = 0.5$, $r_0 = 0.5$, $d_0 = 1.0$, and $d_1 = 1$. For the low variance setting (the first row of plots), we use $r_1 \sim \mathcal{N}(1.5, 0.033)$. For the high variance setting (the second row of plots), we use $r_1 \sim \mathcal{N}(1.5, 0.1)$. The sample size n varies from 500 to 500×2^{15} . We record the mean relative errors of estimators $\hat{\lambda}_0^{(n)}$, $\hat{\lambda}_1^{(n)}$, $\hat{\alpha}^{(n)}$, and $\hat{r}_1^{(n)}$, as well as the observable quantities γ_n , $I_n(\gamma_n)$, $R_n(\gamma_n)$, and $Z_0^n(\gamma_n)$

clones have random fitness through a simulation experiment. Specifically, we modify the simulation experiments from Sect. 4.1 to assign random fitness levels to the resistant cells. The birth rate of each newly mutated resistant cell is set to follow a Gaussian distribution, $r_1 \sim \mathcal{N}(1.5, 0.033)$ for a low variance setting and $r_1 \sim \mathcal{N}(1.5, 0.1)$ for a high variance setting. The rest of the simulation process remains unchanged. We then record the observable quantities and calculate the estimators as before. The performance of the estimators and the observable quantities are presented in Fig. 6.

We observe that in both low variance and high variance settings, the performance of the estimators $\hat{\lambda}_1^{(n)}$ and $\hat{r}_1^{(n)}$, as well as the observable quantity $R_n(\gamma_n)$, is significantly

degraded, while the other estimators still converge well. To justify this phenomenon, note that the random fitness effect does not affect the birth, death, and mutation events of sensitive cells and has a non-significant effect on the overall growth of the mutant population. Therefore, our theoretical results, Theorems 1 and 2 (regarding γ_n , $I_n(\gamma_n)$ and $Z_0^n(\gamma_n)$), would not be significantly affected. However, the increased variance in clone sizes greatly affects our estimation of the Simpson Index. Furthermore, from Theorem 4, we know the estimators $\hat{\lambda}_1^{(n)}$ and $\hat{r}_1^{(n)}$ depend on the estimation of the Simpson Index, whereas $\hat{\lambda}_0^{(n)}$ and $\hat{\alpha}^{(n)}$ do not. This explains why only $\hat{\lambda}_1^{(n)}$ and $\hat{r}_1^{(n)}$ are significantly affected.

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Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- Athreya KB, Ney PE (1972) Branching processes. Springer, New York
- Avanzini S, Antal T (2019) Cancer recurrence times from a branching process model. *PLoS Comput Biol* 15(11):e1007423
- Bozic I, Nowak MA (2014) Timing and heterogeneity of mutations associated with drug resistance in metastatic cancers. *Proc Natl Acad Sci* 111(45):15964–15968
- Bozic I, Reiter JG, Allen B, Antal T, Chatterjee K, Shah P, Moon YS, Yaqubie A, Kelly N, Le DT et al (2013) Evolutionary dynamics of cancer in response to targeted combination therapy. *Elife* 2:e00747
- Brouard V (2023) Genetic composition of supercritical branching populations under power law mutation rates. *arXiv preprint arXiv:2309.12055*
- Chaft JE, Oxnard GR, Sima CS, Kris MG, Miller VA, Riely GJ (2011) Disease flare after tyrosine kinase inhibitor discontinuation in patients with EGFR-mutant lung cancer and acquired resistance to erlotinib or gefitinib: implications for clinical trial design. *Clin Cancer Res* 17(19):6298–6303
- Cheek D, Antal T (2020) Genetic composition of an exponentially growing cell population. *Stoch Process Appl* 130(11):6580–6624
- Cibulskis K, Lawrence MS, Carter SL, Sivachenko A, Jaffe D, Sougnez C, Gabriel S, Meyerson M, Lander ES, Getz G (2013) Sensitive detection of somatic point mutations in impure and heterogeneous cancer samples. *Nat Biotechnol* 31(3):213–219
- Dentro SC, Wedge DC, Van Loo P (2017) Principles of reconstructing the subclonal architecture of cancers. *Cold Spring Harb Perspect Med* 7(8):a026625
- Ding L, Ley TJ, Larson DE, Miller CA, Koboldt DC, Welch JS, Ritchey JK, Young MA, Lamprecht T, McLellan MD et al (2012) Clonal evolution in relapsed acute myeloid Leukaemia revealed by whole-genome sequencing. *Nature* 481(7382):506–510
- Duesberg P, Stindl R, Hehlmann R (2000) Explaining the high mutation rates of cancer cells to drug and multidrug resistance by chromosome reassortments that are catalyzed by aneuploidy. *Proc Natl Acad Sci* 97(26):14295–14300
- Durrett R, Foo J, Leder K (2016) Spatial Moran models, ii: cancer initiation in spatially structured tissue. *J Math Biol* 72:1369–1400
- Durrett R, Foo J, Leder K, Mayberry J, Michor F (2011) Intratumor heterogeneity in evolutionary models of tumor progression. *Genetics* 188(2):461–477

- Elowitz MB, Levine AJ, Siggia ED, Swain PS (2002) Stochastic gene expression in a single cell. *Science* 297(5584):1183–1186
- Feinerma O, Veiga J, Dorfman JR, Germain RN, Altan-Bonnet G (2008) Variability and robustness in t cell activation from regulated heterogeneity in protein levels. *Science* 321(5892):1081–1084
- Foo J, Leder K (2013) Dynamics of cancer recurrence. *Ann Appl Probab* 23(4):1437–1468
- Foo J, Leder K, Mumenthaler SM (2013) Cancer as a moving target: understanding the composition and rebound growth kinetics of recurrent tumors. *Evol Appl* 6(1):54–69
- Foo J, Leder K, Zhu J (2014) Escape times for branching processes with random mutational fitness effects. *Stoch Process Appl* 124(11):3661–3697
- Gatenby RA, Silva AS, Gillies RJ, Frieden BR (2009) Adaptive therapy. *Can Res* 69(11):4894–4903
- Giuliano AE, Dale PS, Turner RR, Morton DL, Evans SW, Krasne DL (1995) Improved axillary staging of breast cancer with sentinel lymphadenectomy. *Ann Surg* 222(3):394
- Gunnarsson EB, Foo J, Leder K (2023) Statistical inference of the rates of cell proliferation and phenotypic switching in cancer. *J Theor Biol* 568:111497
- Gunnarsson EB, Leder K, Foo J (2021) Exact site frequency spectra of neutrally evolving tumors: a transition between power laws reveals a signature of cell viability. *Theor Popul Biol* 142:67–90
- Gunnarsson EB, Leder K, Zhang X (2023) Limit theorems for the site frequency spectrum of neutral mutations in an exponentially growing population. *arXiv preprint* [arXiv:2307.03346](https://arxiv.org/abs/2307.03346)
- Ha G, Roth A, Khattri J, Ho J, Yap D, Prentice LM, Melnyk N, McPherson A, Bashashati A, Laks E, Biele J, Ding J, Le A, Rosner J, Shumansky K, Marra MA, Gilks CB, Huntsman DG, McAlpine JN, Aparicio S, Shah SP (2014) TITAN: Inference of copy number architectures in clonal cell populations from tumor whole-genome sequence data. *Genome Res* 24(11):1881–1893
- Hanagal P, Leder K, Wang Z (2022) Large deviations of cancer recurrence timing. *Stoch Process Appl* 147:1–50
- Housman G et al (2014) Drug resistance in cancer: an overview. *Cancers* 6(3):1769–1792
- Hunter C et al (2006) A hypermutation phenotype and somatic msh6 mutations in recurrent human malignant gliomas after alkylator chemotherapy. *Can Res* 66(8):3987–3991
- Ichihara E, Lovly CM (2015) Shades of t790m: intratumor heterogeneity in EGFR-mutant lung cancer. *Cancer Discov* 5(7):694–696
- Ilic N, Utermark T, Widlund HR, Roberts TM (2011) Pi3k-targeted therapy can be evaded by gene amplification along the MYC-eukaryotic translation initiation factor 4e (eif4e) axis. *Proc Natl Acad Sci* 108(37):E699–E708
- Iwasa Y, Nowak MA, Michor F (2006) Evolution of resistance during clonal expansion. *Genetics* 172(4):2557–2566
- Jagers P, Klebaner FC, Sagitov S (2007) Markovian paths to extinction. *Adv Appl Probab* 39(2):569–587
- Jasmine LK, Leder K (2012) Rare events in cancer recurrence timing. In: *Proceedings of the 2012 winter simulation conference (WSC)*, pp 1–10
- Komarova NL, Wodarz D (2005) Drug resistance in cancer: principles of emergence and prevention. *Proc Natl Acad Sci* 102(27):9714–9719
- Leder K, Wang Z (2021) Clonal diversity at cancer recurrence. *arXiv preprint* [arXiv:2108.13472](https://arxiv.org/abs/2108.13472)
- Lee ND, Bozic I (2022) Inferring parameters of cancer evolution in chronic lymphocytic leukemia. *PLoS Comput Biol* 18(11):e1010677
- Li A, Kibby D, Foo J (2023) A comparison of mutation and amplification-driven resistance mechanisms and their impacts on tumor recurrence. *J Math Biol* 87(4):59
- Liang H, Sha N (2004) Modeling antitumor activity by using a non-linear mixed-effects model. *Math Biosci* 189(1):61–73
- Miller CA, White BS, Dees ND, Griffith M, Welch JS, Griffith OL, Vij R, Tomasson MH, Graubert TA, Walter MJ, Ellis MJ, Schierding W, DiPersio JF, Ley TJ, Mardis ER, Wilson RK, Ding L (2014) SciClone: inferring clonal architecture and tracking the spatial and temporal patterns of tumor evolution. *PLoS Comput Biol* 10(8):e1003665
- Narod SA (2012) Disappearing breast cancers
- Nicholson MD, Cheek D, Antal T (2023) Sequential mutations in exponentially growing populations. *PLoS Comput Biol* 19(7):e1011289
- Nussinov R, Tsai C-J, Jang H (2021) Anticancer drug resistance: an update and perspective. *Drug Resist Updates* 59:100796
- Piotrowska Z, Niederst MJ, Karlovich CA, Wakelee HA, Neal JW, Mino-Kenudson M, Fulton L, Hata AN, Lockerman EL, Kalsy A et al (2015) Heterogeneity underlies the emergence of EGFR T790 wild-type

- clones following treatment of t790m-positive cancers with a third-generation egfr inhibitor. *Cancer Discov* 5(7):713–722
- Roth A, Khattra J, Yap D, Wan A, Laks E, Biele J, Ha G, Aparicio S, Bouchard-Côté A, Shah SP (2014) Pyclone: statistical inference of clonal population structure in cancer. *Nat Methods* 11:396–398
- Salichos L, Meyerson W, Warrell J, Gerstein M (2020) Estimating growth patterns and driver effects in tumor evolution from individual samples. *Nat Commun* 11(1):732
- Stead LF, Sutton KM, Taylor GR, Quirke P, Rabbitts P (2013) Accurately identifying low-allelic fraction variants in single samples with next-generation sequencing: applications in tumor subclone resolution. *Hum Mutat* 34:1432–1438
- Strobl MAR, Gallaher J, West J, Robertson-Tessi M, Maini PK, Anderson ARA (2022) Spatial structure impacts adaptive therapy by shaping intra-tumoral competition. *Commun Med* 2(1):46
- Suda K, Onozato R, Yatabe Y, Mitsudomi T (2009) Egfr t790m mutation: a double role in lung cancer cell survival? *J Thorac Oncol* 4(1):1–4
- Sze W-M, Lee AWM, Yau T-K, Yeung RMW, Lau K-Y, Leung SKC, Hung AWM, Lee MCH, Chappell R, Chan K (2004) Primary tumor volume of nasopharyngeal carcinoma: prognostic significance for local control. *Int J Radiat Oncol Biol Phys* 59(1):21–27
- VanLoo P, Nordgard SH, Lingjærde OC, Russnes HG, Rye IH, Sun W, Weigman VJ, Marynen P, Zetterberg A, Naume B, Perou CM, Børresen-Dale A-L, Kristensen VN (2010) Allele-specific copy number analysis of tumors. *Proc Natl Acad Sci* 107(39):16910–16915
- Wang S, Tsui ST, Liu C, Song Y, Liu D (2016) Egfr c797s mutation mediates resistance to third-generation inhibitors in t790m-positive non-small cell lung cancer. *J Hematol Oncol* 9:1–5
- Werner B, Case J, Williams MJ, Chkhaidze K, Temko D, Fernández-Mateos J, Cresswell GD, Nichol D, Cross W, Spiteri I et al (2020) Measuring single cell divisions in human tissues from multi-region sequencing data. *Nat Commun* 11(1):1035

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